Nebico Oil Retention Project (3.2)

Coolmay L10S Ladder program
TK8050 x 3, TK8100 x 1

Modifications:

1. Pointer Functions to shorten bulk calculation
2. Revised Core logic (Mode Scheduler)
3. Revised Oil Distribution
4. Clean Segmentation

Prepared By: Sambridda Ranabhat (Architect & Programmer)

09/09/2026
Mechatronics Engineering .pvt .ltd

Data Name : Memory Capacity
Memory Capacity

[Memory Capacity]
16000

[Program Capacity]
16000 Steps

[Comments Capacity]
0 Block 0 Points

[File Register Capacity]
0 Block 0 Points

[Special Function Block Setting]
0 Block

[Positioning]
0 Block

[Built-in CC-Link/LT Setting]
0 Block

Data Name : PLC System
PLC System

[Battery Less Mode]
OFF

[MODEM Initialized]
None

[RUN Terminal Input]
None

[Device]

	Sym.	Dig.	Points	Start	End	Latch Start	End	Latch Setting Range
Supplemental Relay	M	10	7680	0	7679	500	1023	0 - 1023
State	S	10	4096	0	4095	500	999	0 - 999
Timer	T	10	512	0	511			
Counter(16bit)	C	10	200	0	199	100	199	0 - 199
Counter(32bit)	C	10	56	200	255	220	255	200 - 255
Data Register	D	10	8000	0	7999	200	511	0 - 511
Extended Register	R	10	32768	0	32767			

CC-Link Setting

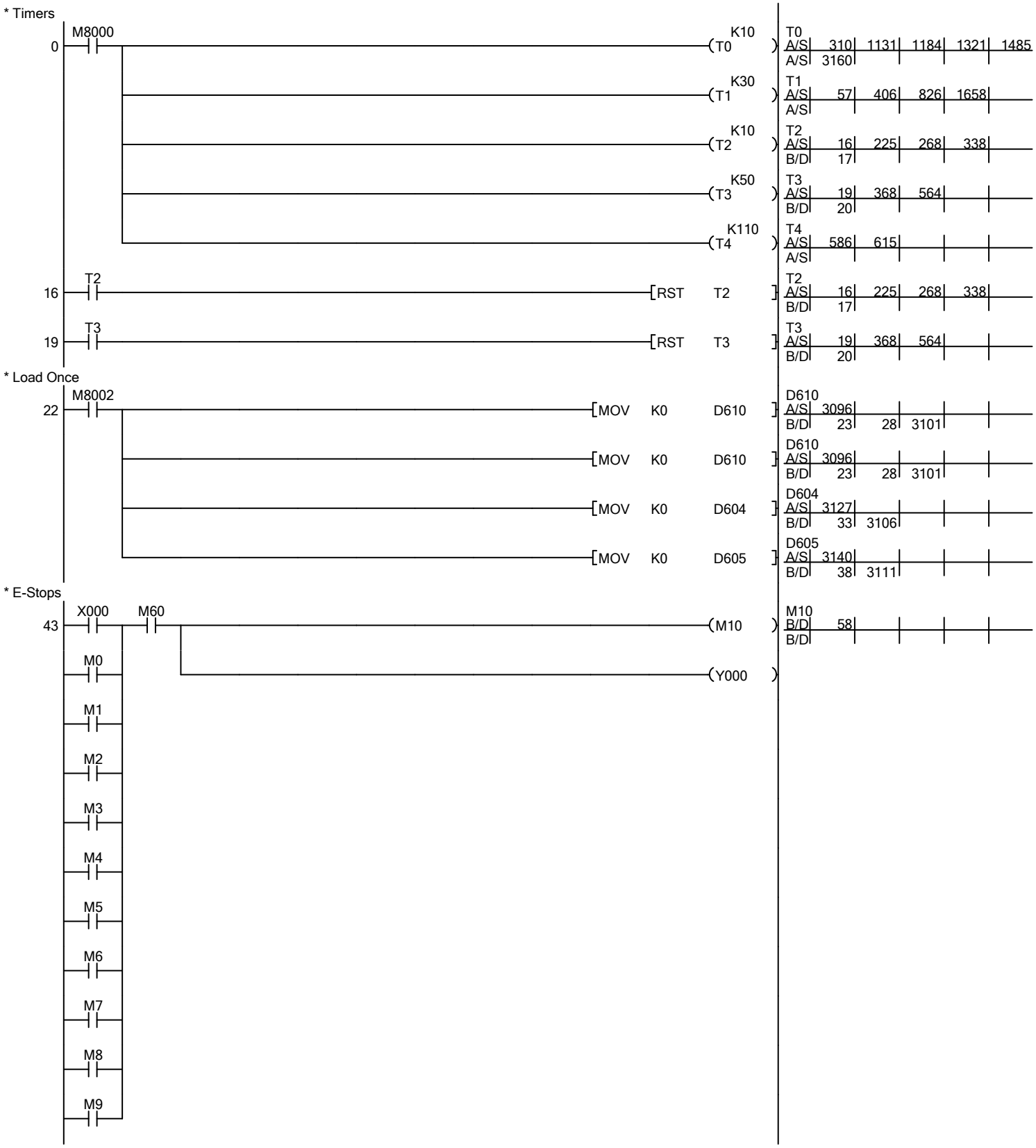
[CC-Link Setting]

Connection Block Not Set

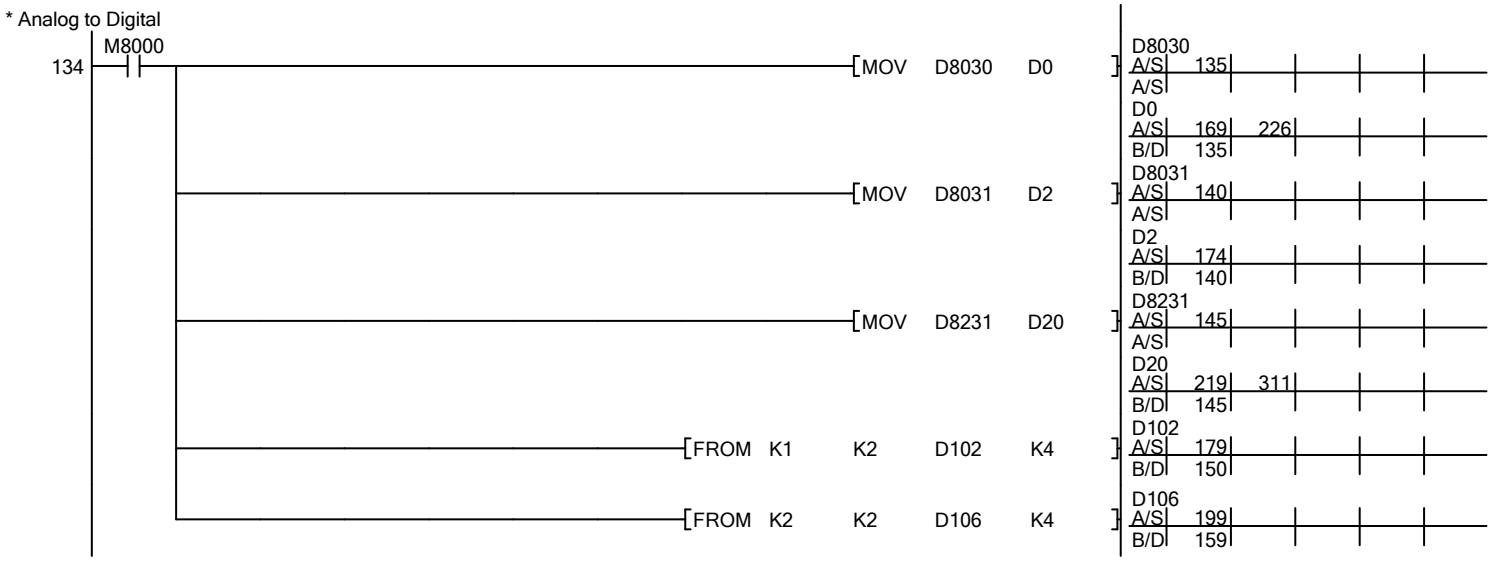
Data Name : CC-Link
CC-Link Setting

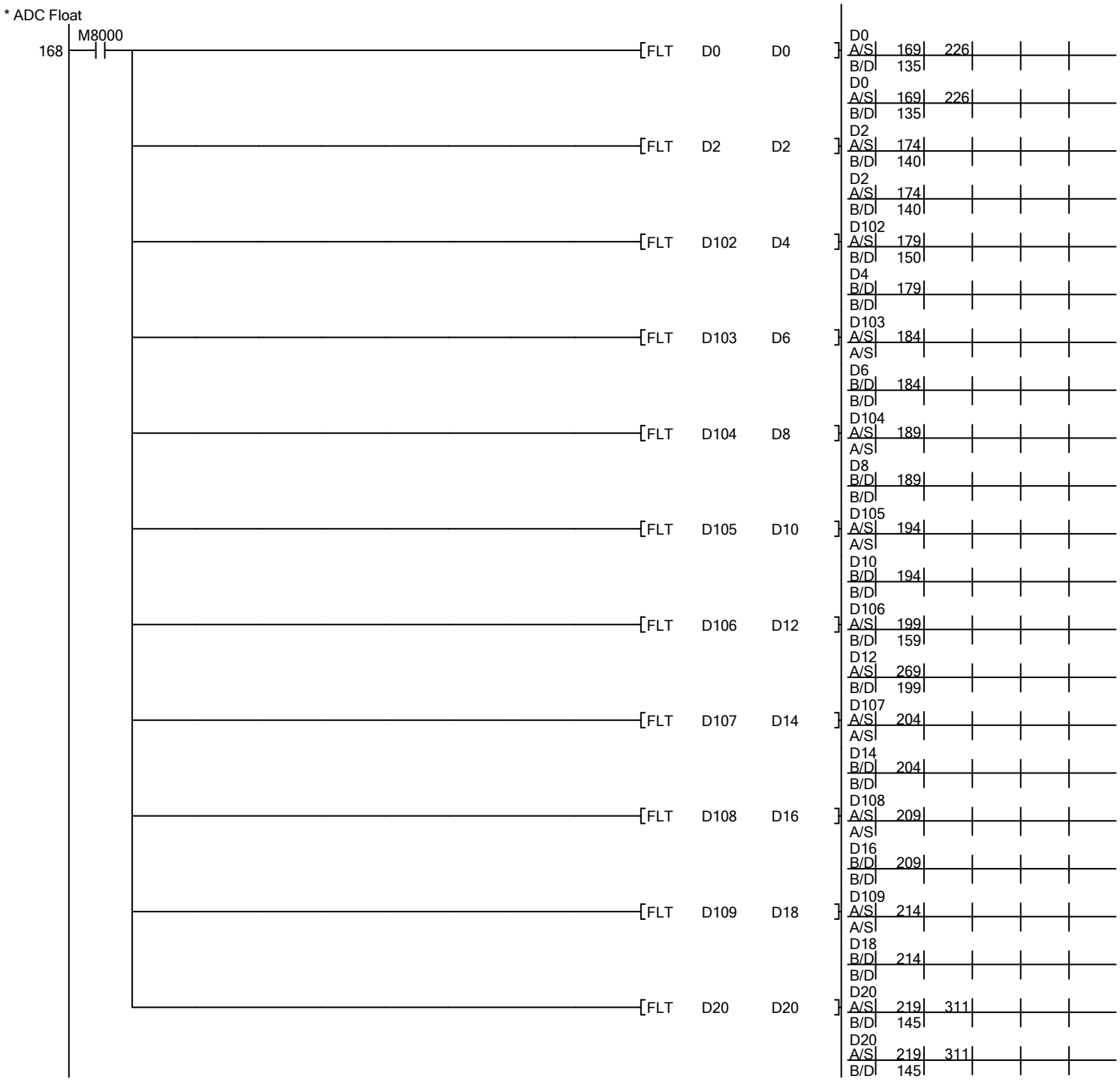
[CC-Link Setting]

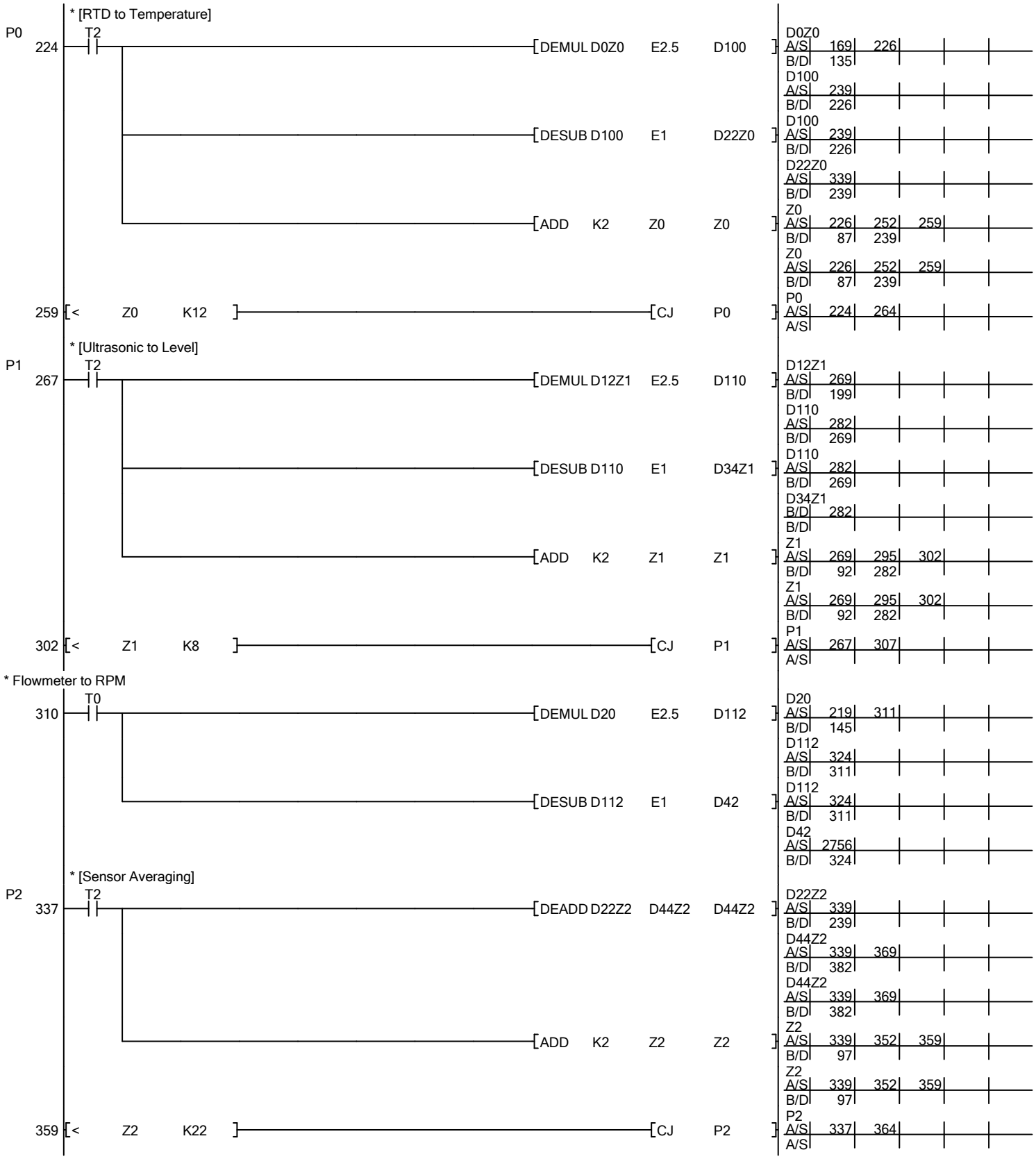
	Setting Contents
Special Function Block No.	-
Type	-
Master Station Data Link Type	Start Parameter by BFM
Mode	-
Total Module Connected	-
Remote Input(RX)	-
Remote Output(RY)	-
Remote Register(RWr)	-
Remote Register(RWw)	-
Ver.2 Remote Input(RX)	-
Ver.2 Remote Output(RY)	-
Ver.2 Remote Register(RWr)	-
Ver.2 Remote Register(RWw)	-
Special Relay(SB)	-
Special Register(SW)	-
Retry Count	-
Automatic Reconnection Station Count	-
Standby Master Station No.	-
PLC Down Select	-
Scan Mode Setting	-
Delay Time Setting	-
Remote Device Station Initial Setting	-
Interrupt Settings	-

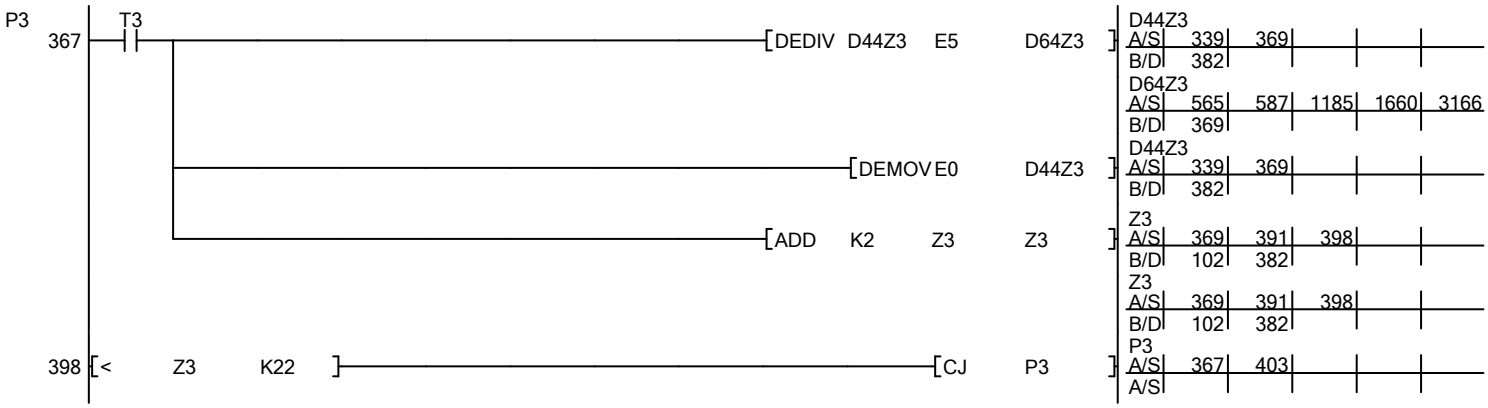

* Timers
0
M8000
K10
T0
T0
A/S 310 1131 1184 1321 1485
A/S 3160
K30
T1
T1
A/S 57 406 826 1658
A/S
K10
T2
T2
A/S 16 225 268 338
B/D 17
K50
T3
T3
A/S 19 368 564
B/D 20
K110
T4
T4
A/S 586 615
A/S
16
T2
RST T2
T2
A/S 16 225 268 338
B/D 17
19
T3
RST T3
T3
A/S 19 368 564
B/D 20
* Load Once
22
M8002
MOV K0 D610
D610
A/S 3096
B/D 23 28 3101
MOV K0 D610
D610
A/S 3096
B/D 23 28 3101
MOV K0 D604
D604
A/S 3127
B/D 33 3106
MOV K0 D605
D605
A/S 3140
B/D 38 3111
* E-Stops
43
X000
M60
M10
M10
B/D 58
B/D
Y000
M0
M1
M2
M3
M4
M5
M6
M7
M8
M9

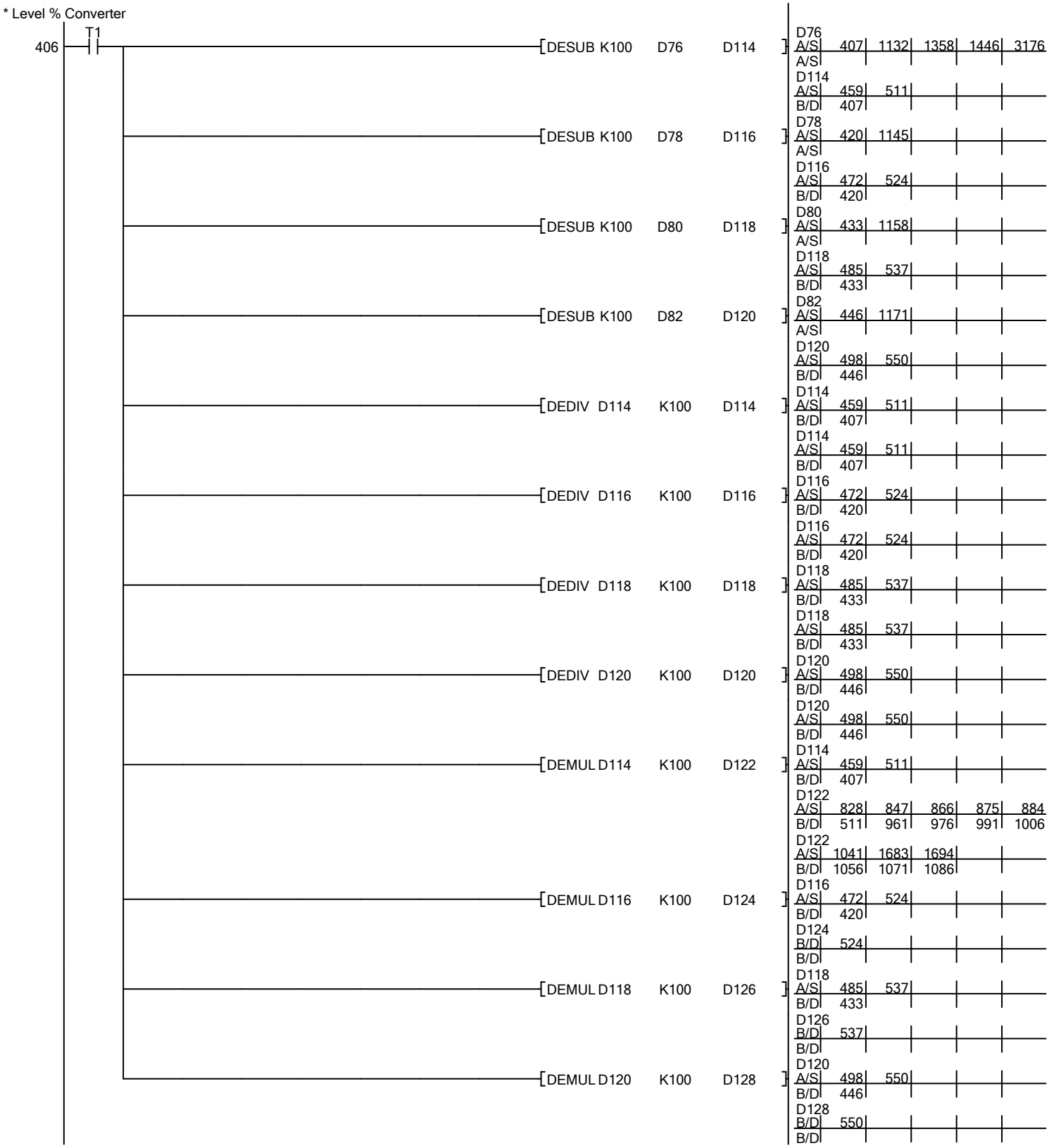

* Automation
57
T1
M10
X001
M11
Y001
X001
M12
M11
A/S 949 1726 1745 1761 2272
A/S 2446 2451 2456 2461 2466
M11
A/S 2470 2474 2478 2482 2486
A/S 2489 2518 2842 2898 2903
M12
A/S 1121 1765 2514 2895 2899
A/S 2933 2961 2964 2996 2999
M12
A/S 3021 3024
A/S
* Declarations
66
M8000
MOV K15 D8114
MOV K0 D550
MOV K0 D304
MOV K0 D303
MOV K0 Z0
MOV K0 Z1
MOV K0 Z2
MOV K0 Z3
MOV K0 Z4
MOV K0 Z5
MOV K97 D8400
MOV H1 D8401
SET M8192
MOV K1 D8397
D8114
B/D 67
B/D
D550
A/S 2652 2685 2718
B/D 72
D304
A/S 1472 2670 2703 2736
B/D 77 2679 2712 2745
D303
A/S 2560 2598 2636
B/D 82 2568 2606 2644
Z0
A/S 226 252 259
B/D 87 239
Z1
A/S 269 295 302
B/D 92 282
Z2
A/S 339 352 359
B/D 97
Z3
A/S 369 391 398
B/D 102 382
Z4
A/S 565 570 577
B/D 107
Z5
A/S 587 600 607
B/D 112
D8400
B/D 117
B/D
D8401
B/D 122
B/D
M8192
A/S 127
A/S
D8397
B/D 129
B/D

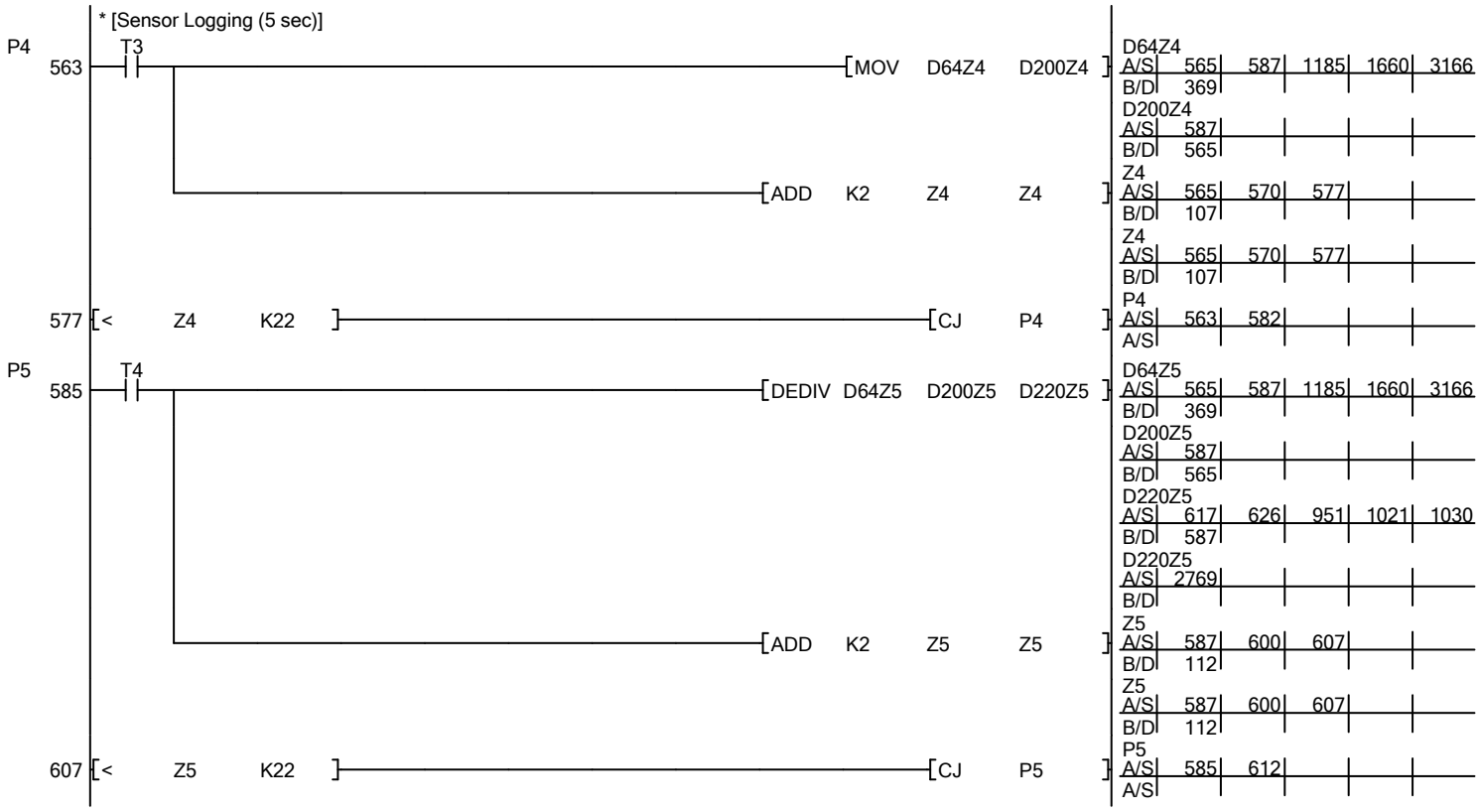

* Analog to Digital
M8000
134
MOV D8030 D0
MOV D8031 D2
MOV D8231 D20
FROM K1 K2 D102 K4
FROM K2 K2 D106 K4
D8030
A/S 135
A/S
D0
A/S 169 226
B/D 135
D8031
A/S 140
A/S
D2
A/S 174
B/D 140
D8231
A/S 145
A/S
D20
A/S 219 311
B/D 145
D102
A/S 179
B/D 150
D106
A/S 199
B/D 159

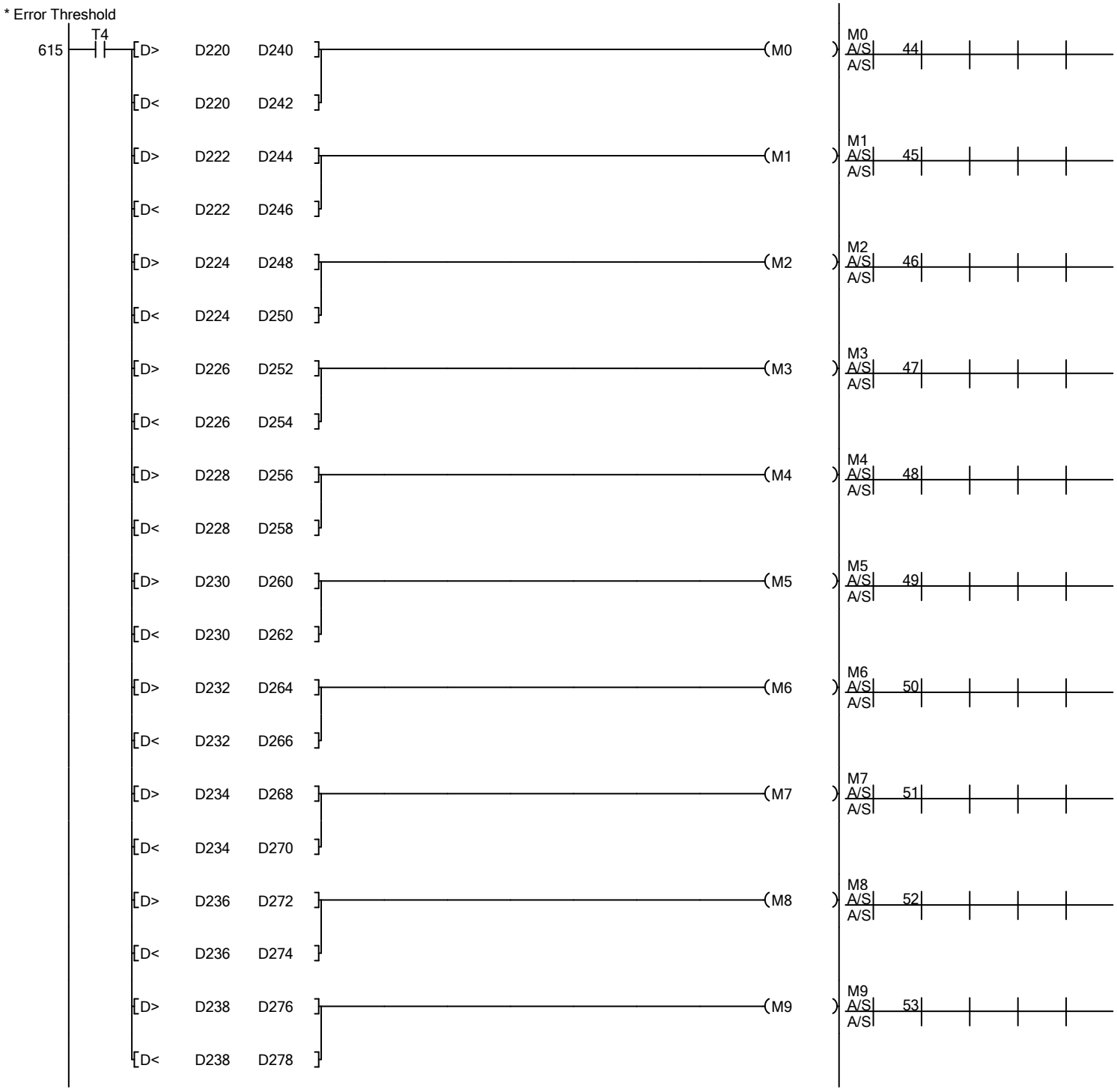

* ADC Float
M8000
168
FLT D0 D0
FLT D2 D2
FLT D102 D4
FLT D103 D6
FLT D104 D8
FLT D105 D10
FLT D106 D12
FLT D107 D14
FLT D108 D16
FLT D109 D18
FLT D20 D20
D0 A/S 169 226
B/D 135
D0 A/S 169 226
B/D 135
D2 A/S 174
B/D 140
D2 A/S 174
B/D 140
D102 A/S 179
B/D 150
D4 B/D 179
B/D
D103 A/S 184
A/S
D6 B/D 184
B/D
D104 A/S 189
A/S
D8 B/D 189
B/D
D105 A/S 194
A/S
D10 B/D 194
B/D
D106 A/S 199
B/D 159
D12 A/S 269
B/D 199
D107 A/S 204
A/S
D14 B/D 204
B/D
D108 A/S 209
A/S
D16 B/D 209
B/D
D109 A/S 214
A/S
D18 B/D 214
B/D
D20 A/S 219 311
B/D 145
D20 A/S 219 311
B/D 145


* [RTD to Temperature]
P0
T2
224
DEMUL D0Z0 E2.5 D100
DESUB D100 E1 D22Z0
ADD K2 Z0 Z0
259 < Z0 K12
CJ P0
* [Ultrasonic to Level]
P1
T2
267
DEMUL D12Z1 E2.5 D110
DESUB D110 E1 D34Z1
ADD K2 Z1 Z1
302 < Z1 K8
CJ P1
* Flowmeter to RPM
T0
310
DEMUL D20 E2.5 D112
DESUB D112 E1 D42
* [Sensor Averaging]
P2
T2
337
DEADD D22Z2 D44Z2 D44Z2
ADD K2 Z2 Z2
359 < Z2 K22
CJ P2

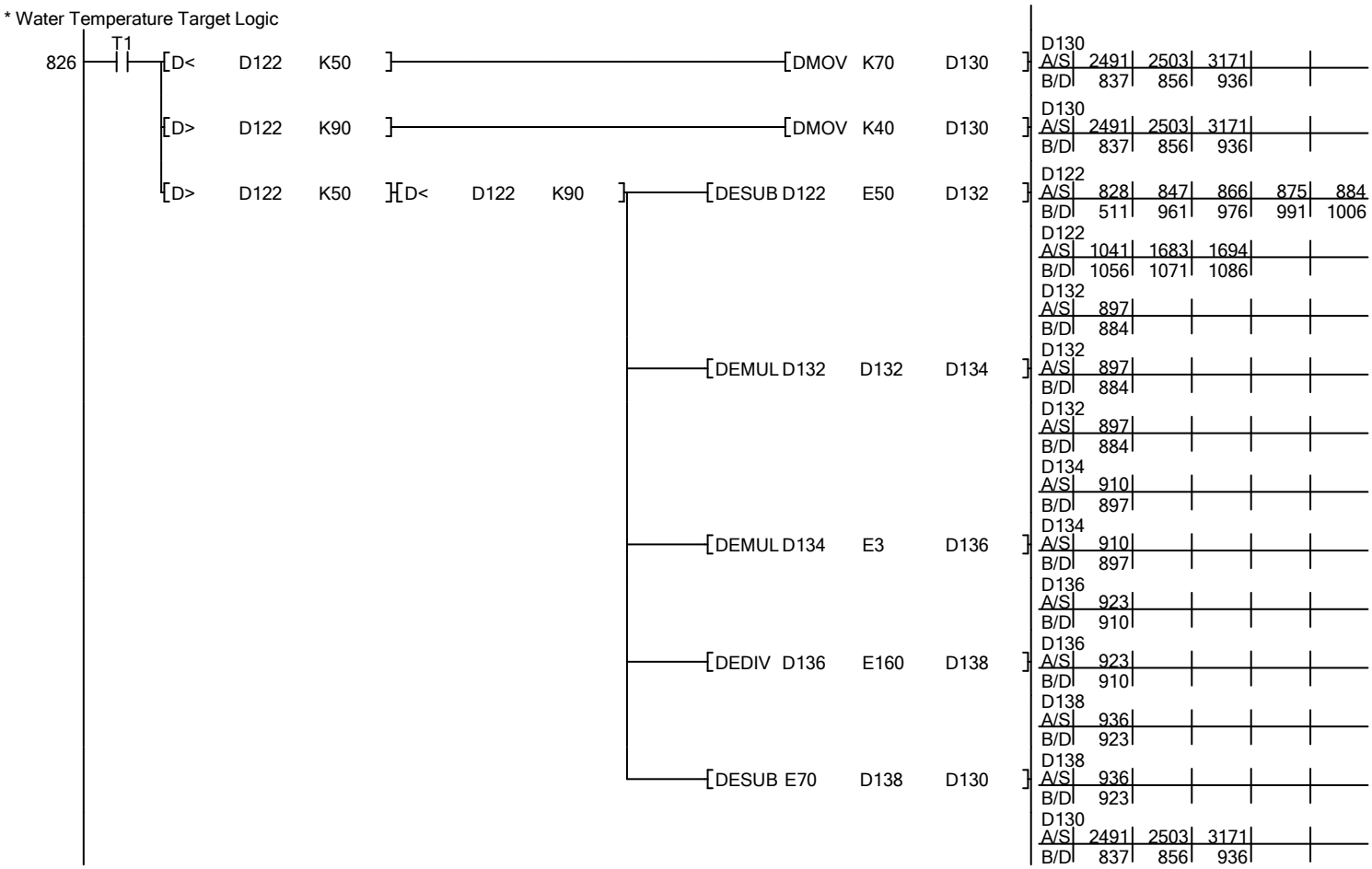

P3
T3
367
DEDIV D44Z3 E5 D64Z3
DEMOV E0 D44Z3
ADD K2 Z3 Z3
398
< Z3 K22
CJ P3
D44Z3
A/S 339 369
B/D 382
D64Z3
A/S 565 587 1185 1660 3166
B/D 369
D44Z3
A/S 339 369
B/D 382
Z3
A/S 369 391 398
B/D 102 382
Z3
A/S 369 391 398
B/D 102 382
P3
A/S 367 403
A/S

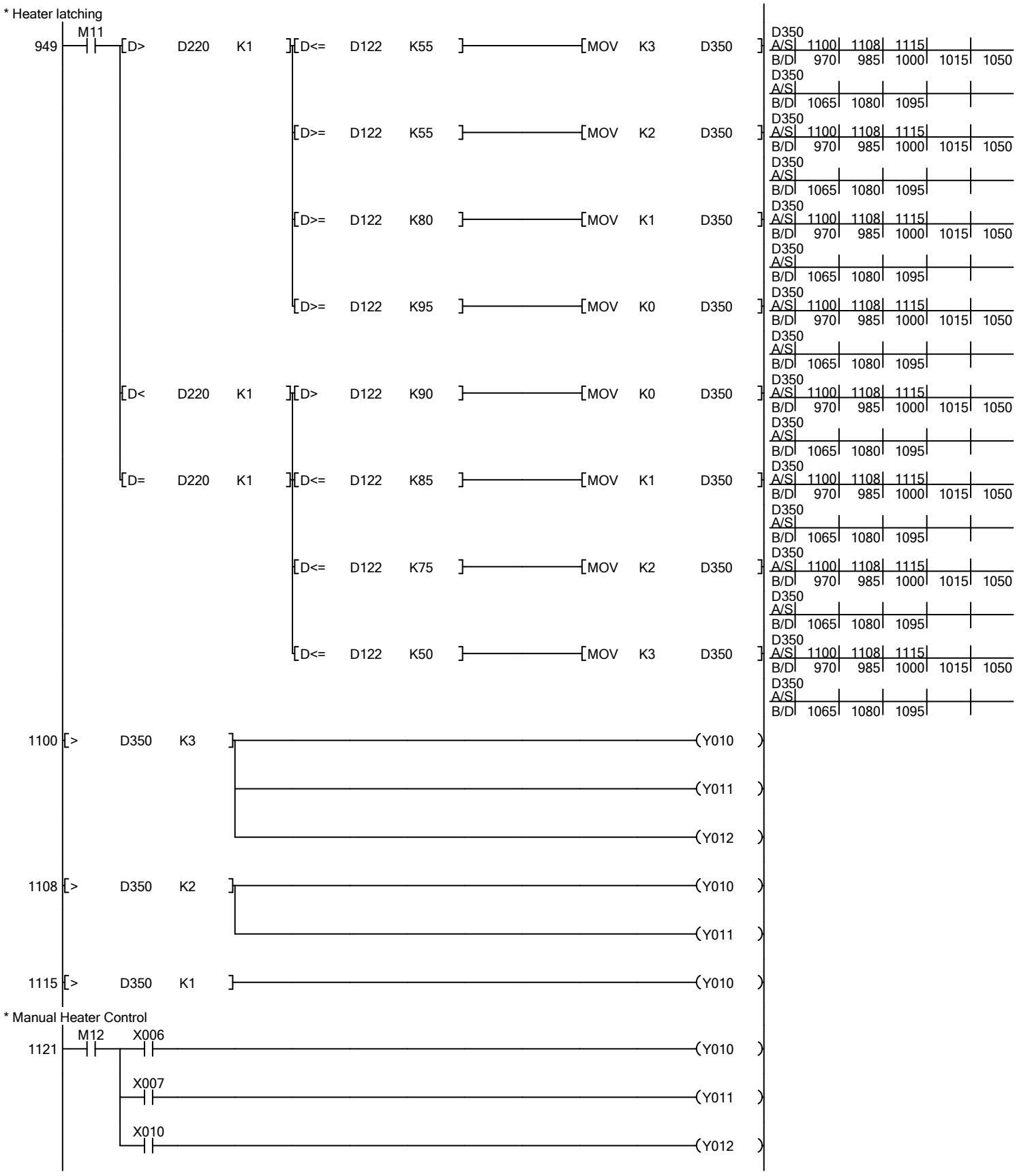

* Level % Converter
406
T1
DESUB K100 D76 D114
DESUB K100 D78 D116
DESUB K100 D80 D118
DESUB K100 D82 D120
DEDIV D114 K100 D114
DEDIV D116 K100 D116
DEDIV D118 K100 D118
DEDIV D120 K100 D120
DEMUL D114 K100 D122
DEMUL D116 K100 D124
DEMUL D118 K100 D126
DEMUL D120 K100 D128
D76 A/S 407 1132 1358 1446 3176
A/S
D114 A/S 459 511 B/D 407
D78 A/S 420 1145
A/S
D116 A/S 472 524 B/D 420
D80 A/S 433 1158
A/S
D118 A/S 485 537 B/D 433
D82 A/S 446 1171
A/S
D120 A/S 498 550 B/D 446
D114 A/S 459 511 B/D 407
D114 A/S 459 511 B/D 407
D116 A/S 472 524 B/D 420
D116 A/S 472 524 B/D 420
D118 A/S 485 537 B/D 433
D118 A/S 485 537 B/D 433
D120 A/S 498 550 B/D 446
D120 A/S 498 550 B/D 446
D114 A/S 459 511 B/D 407
D122 A/S 828 847 866 875 884 B/D 511 961 976 991 1006
D122 A/S 1041 1683 1694 B/D 1056 1071 1086
D116 A/S 472 524 B/D 420
D124 B/D 524
B/D
D118 A/S 485 537 B/D 433
D126 B/D 537
B/D
D120 A/S 498 550 B/D 446
D128 B/D 550
B/D

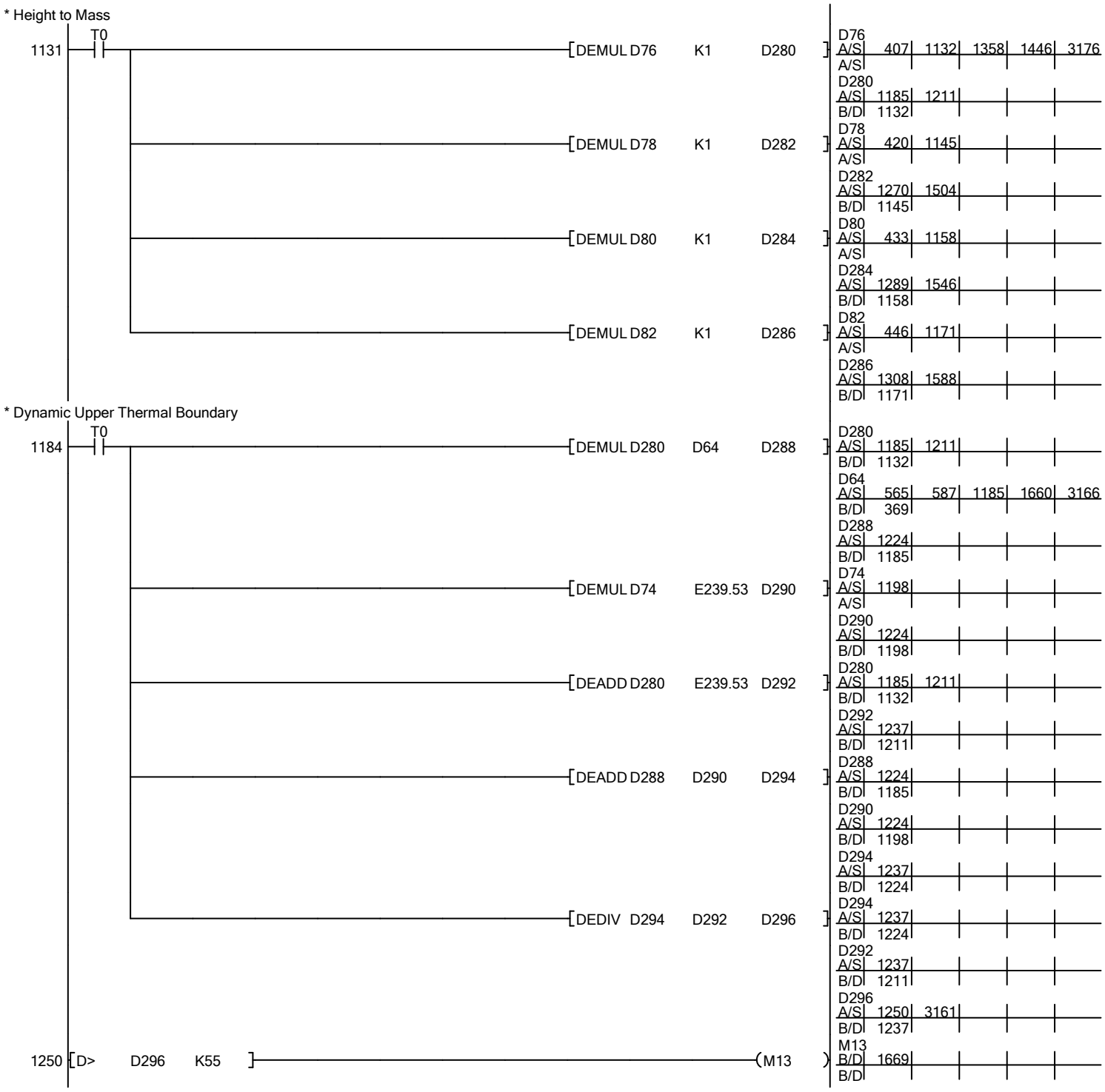

* [Sensor Logging (5 sec)]
P4
T3
563
MOV D64Z4 D200Z4
ADD K2 Z4 Z4
577 < Z4 K22
CJ P4
P5
T4
585
DEDIV D64Z5 D200Z5 D220Z5
ADD K2 Z5 Z5
607 < Z5 K22
CJ P5
D64Z4 A/S 565 587 1185 1660 3166 B/D 369
D200Z4 A/S 587 B/D 565
Z4 A/S 565 570 577 B/D 107
Z4 A/S 565 570 577 B/D 107
P4 A/S 563 582 A/S
D64Z5 A/S 565 587 1185 1660 3166 B/D 369
D200Z5 A/S 587 B/D 565
D220Z5 A/S 617 626 951 1021 1030 B/D 587
D220Z5 A/S 2769 B/D
Z5 A/S 587 600 607 B/D 112
Z5 A/S 587 600 607 B/D 112
P5 A/S 585 612 A/S

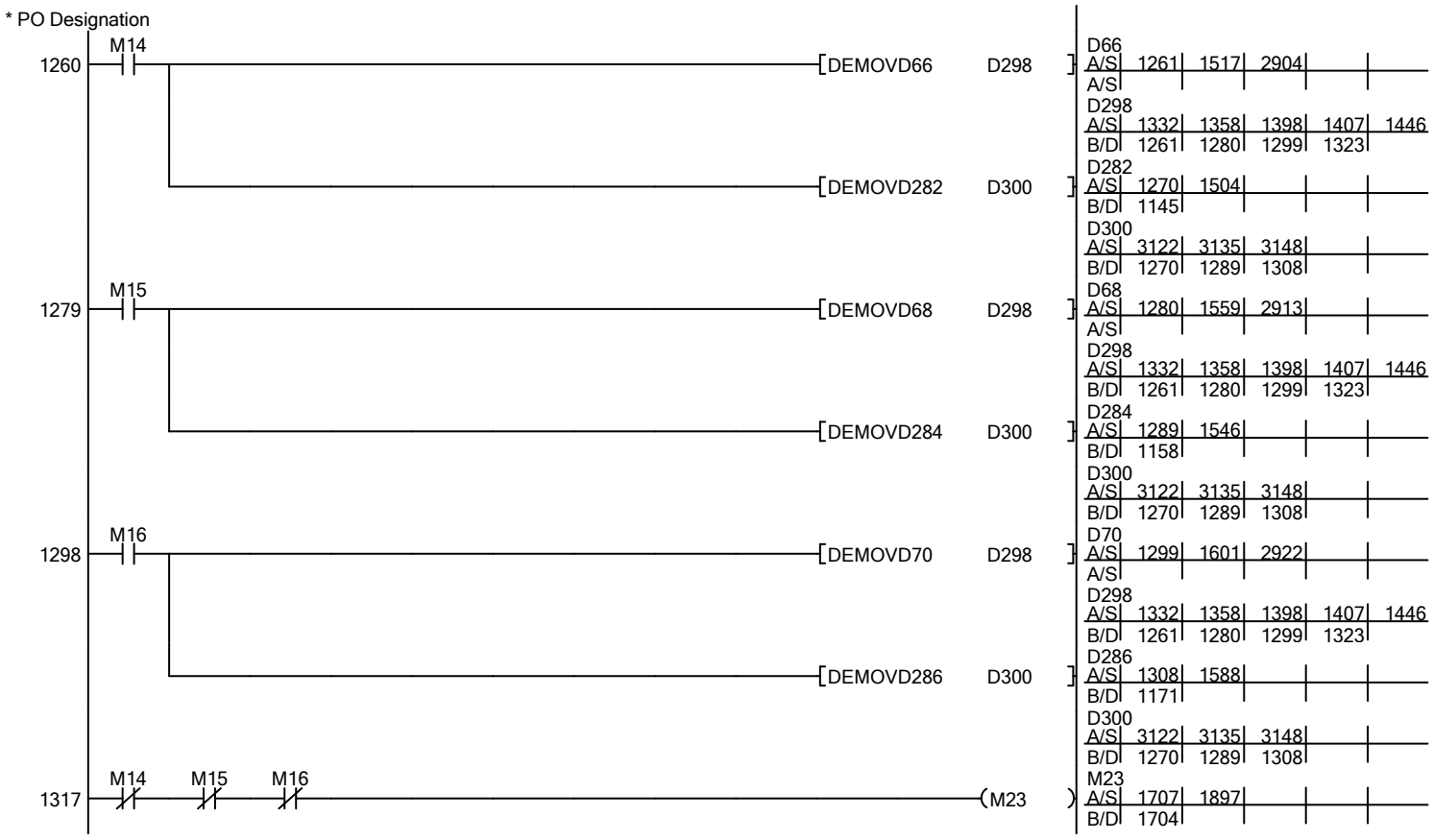

* Error Threshold
615
T4
D> D220 D240
D< D220 D242
D> D222 D244
D< D222 D246
D> D224 D248
D< D224 D250
D> D226 D252
D< D226 D254
D> D228 D256
D< D228 D258
D> D230 D260
D< D230 D262
D> D232 D264
D< D232 D266
D> D234 D268
D< D234 D270
D> D236 D272
D< D236 D274
D> D238 D276
D< D238 D278
M0
M1
M2
M3
M4
M5
M6
M7
M8
M9
M0 A/S 44 A/S
M1 A/S 45 A/S
M2 A/S 46 A/S
M3 A/S 47 A/S
M4 A/S 48 A/S
M5 A/S 49 A/S
M6 A/S 50 A/S
M7 A/S 51 A/S
M8 A/S 52 A/S
M9 A/S 53 A/S

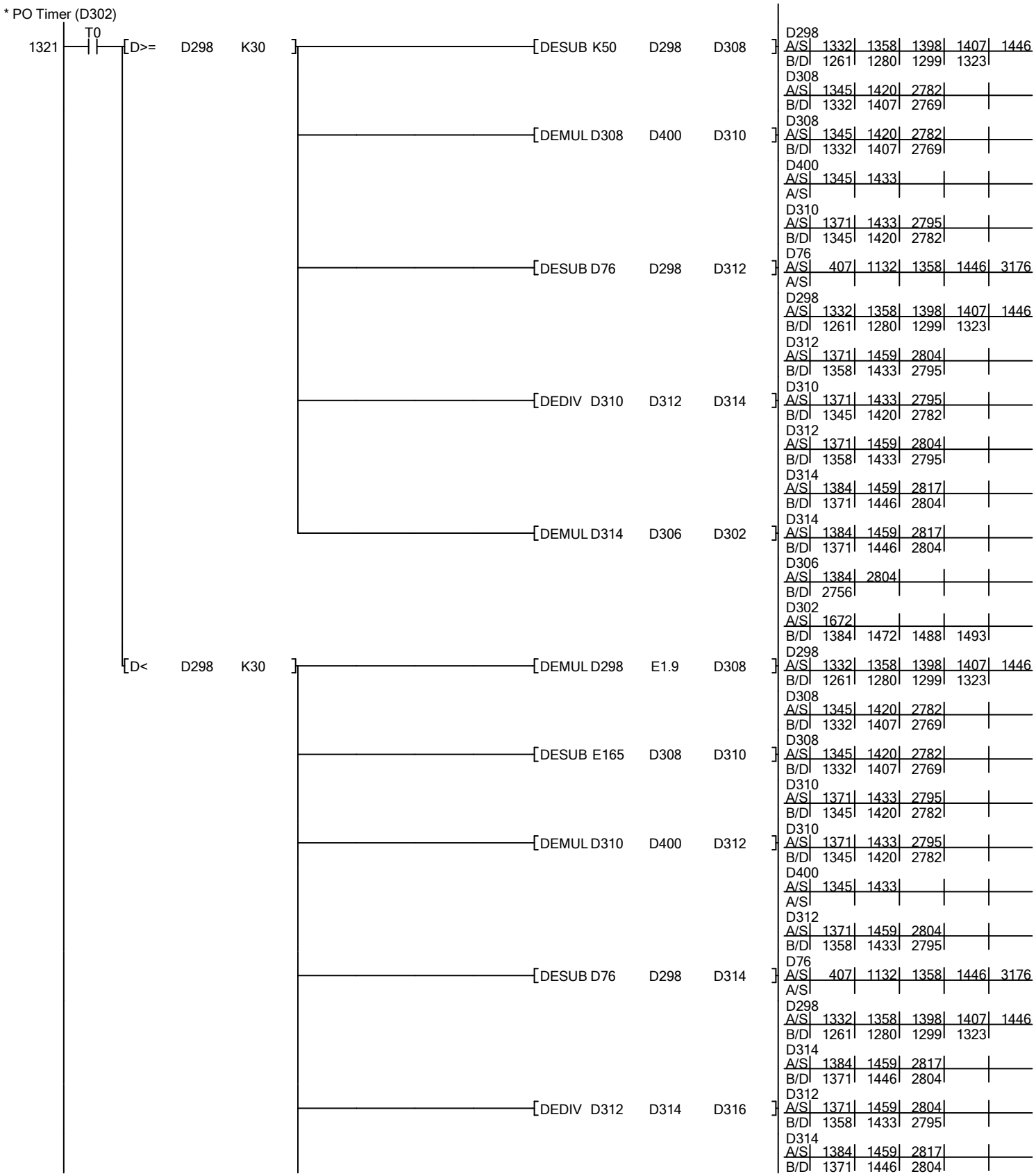
* Water Temperature Target Logic

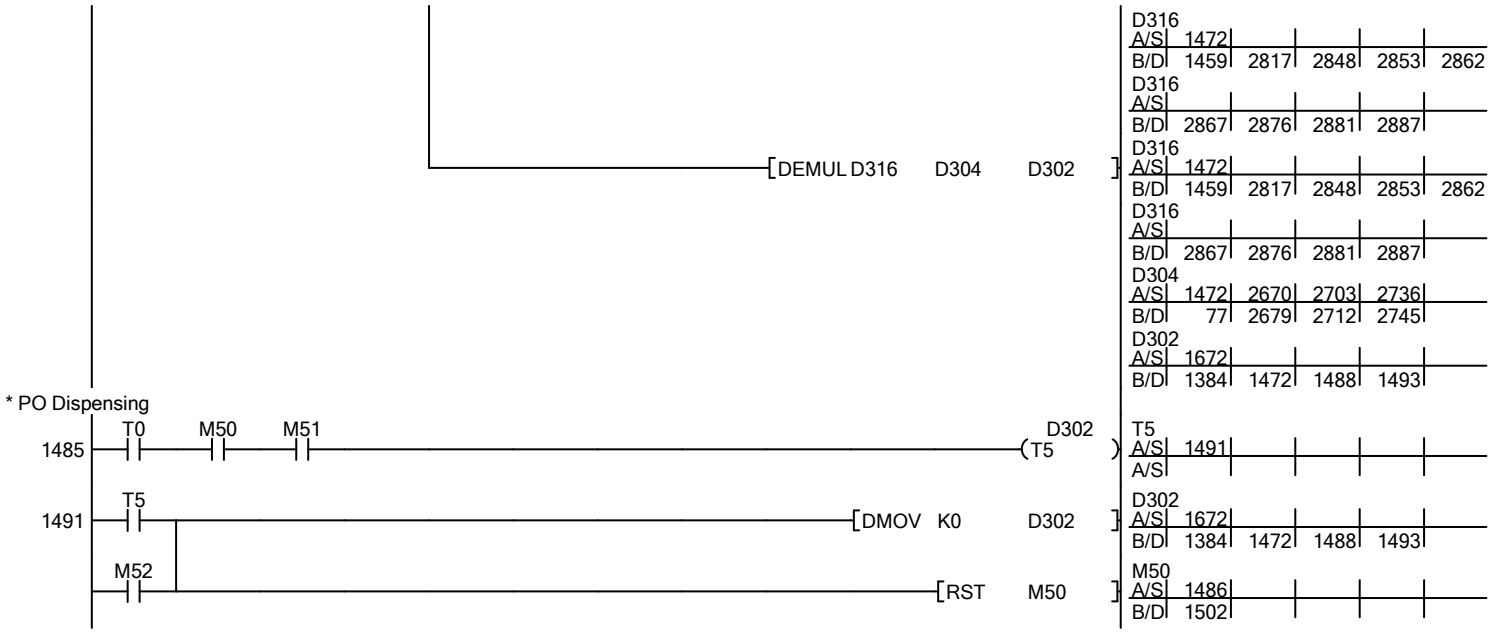


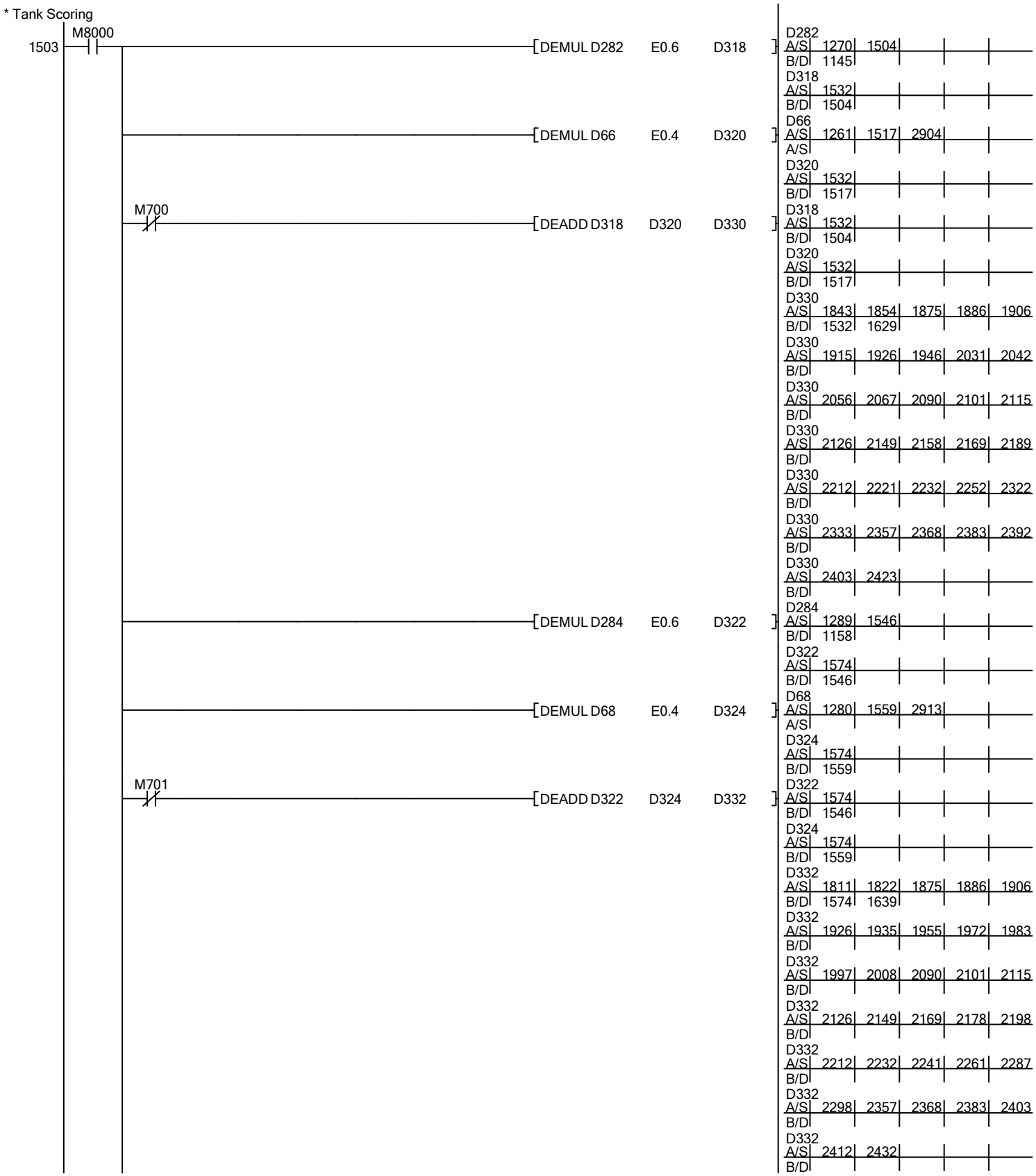

* Heater latching
949 M11 [D> D220 K1] [D<= D122 K55] [MOV K3 D350]
[D>= D122 K55] [MOV K2 D350]
[D>= D122 K80] [MOV K1 D350]
[D>= D122 K95] [MOV K0 D350]
[D< D220 K1] [D> D122 K90] [MOV K0 D350]
[D= D220 K1] [D<= D122 K85] [MOV K1 D350]
[D<= D122 K75] [MOV K2 D350]
[D<= D122 K50] [MOV K3 D350]
D350 A/S 1100 1108 1115
B/D 970 985 1000 1015 1050
D350 A/S
B/D 1065 1080 1095
1100 [> D350 K3] (Y010) (Y011) (Y012)
1108 [> D350 K2] (Y010) (Y011)
1115 [> D350 K1] (Y010)
* Manual Heater Control
1121 M12 X006 (Y010)
X007 (Y011)
X010 (Y012)

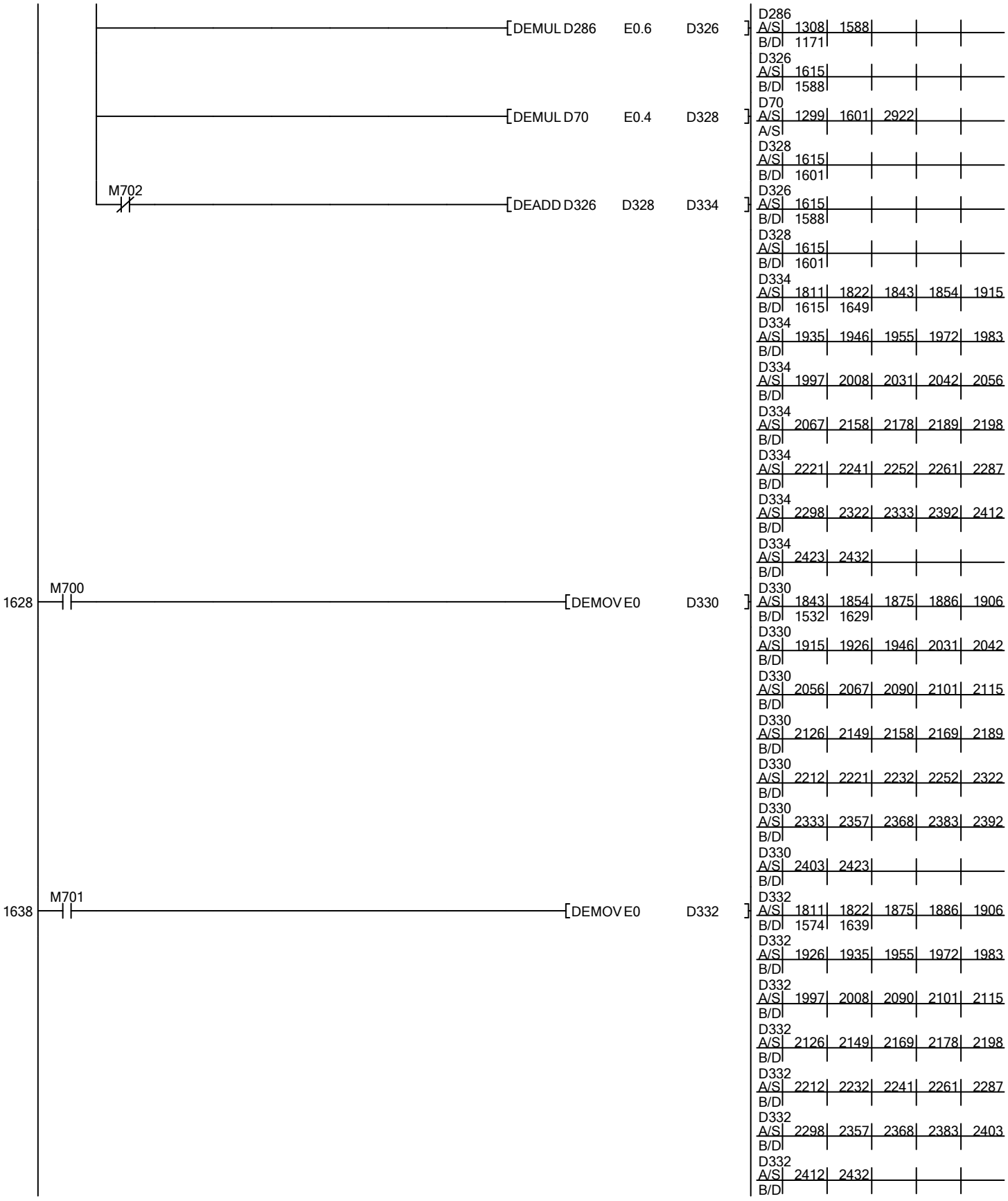

* Height to Mass
1131 T0
DEMUL D76 K1 D280
DEMUL D78 K1 D282
DEMUL D80 K1 D284
DEMUL D82 K1 D286
D76 A/S 407 1132 1358 1446 3176
D280 A/S 1185 1211 B/D 1132
D78 A/S 420 1145
D282 A/S 1270 1504 B/D 1145
D80 A/S 433 1158
D284 A/S 1289 1546 B/D 1158
D82 A/S 446 1171
D286 A/S 1308 1588 B/D 1171
* Dynamic Upper Thermal Boundary
1184 T0
DEMUL D280 D64 D288
DEMUL D74 E239.53 D290
DEADD D280 E239.53 D292
DEADD D288 D290 D294
DEDIV D294 D292 D296
D64 A/S 565 587 1185 1660 3166 B/D 369
D74 A/S 1198
D288 A/S 1224 B/D 1185
D290 A/S 1224 B/D 1198
D292 A/S 1237 B/D 1211
D294 A/S 1237 B/D 1224
D296 A/S 1250 3161 B/D 1237
1250 D> D296 K55
M13
M13 B/D 1669

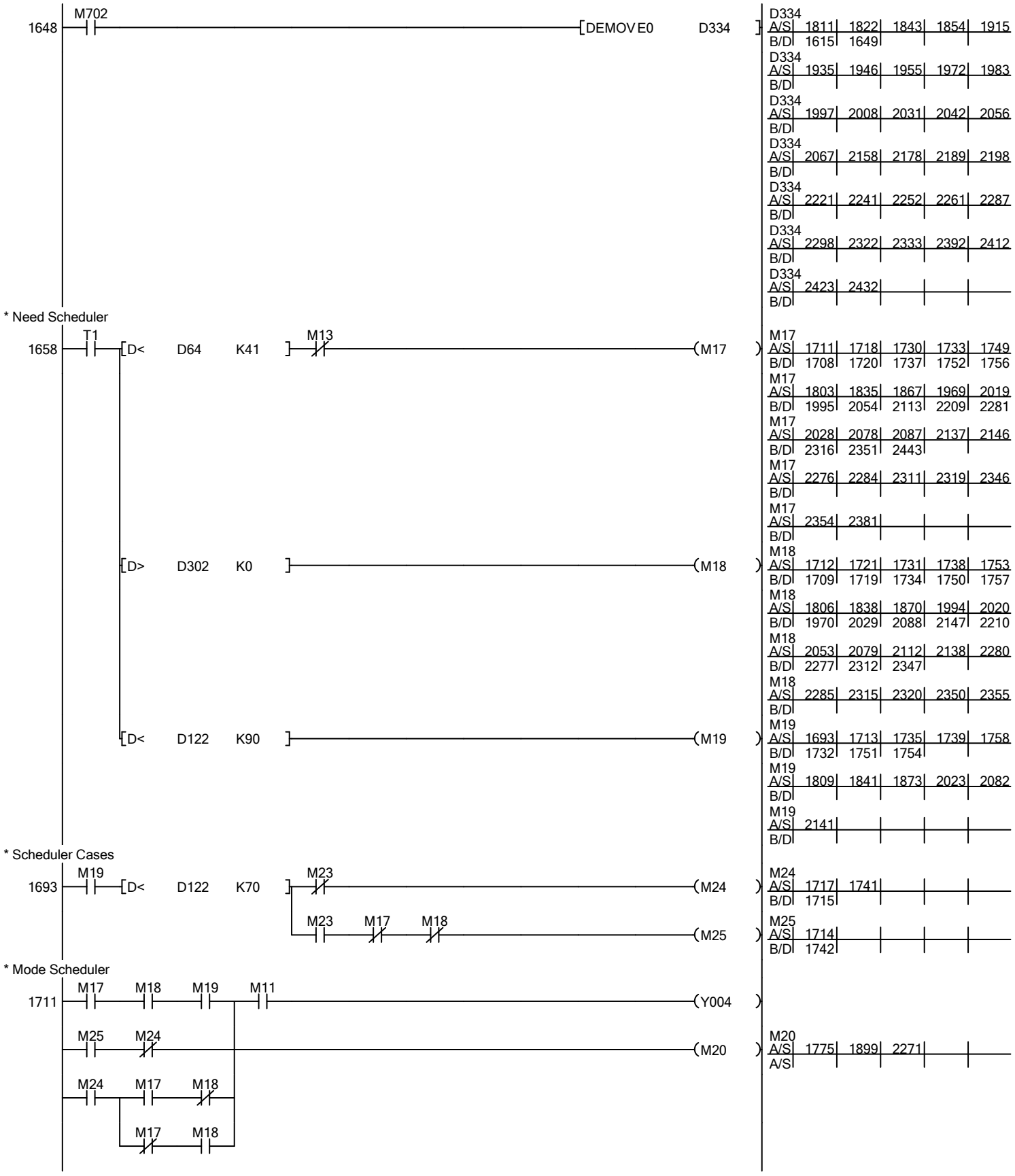

* PO Designation
M14
1260
DEMOVD66 D298
DEMOVD282 D300
M15
1279
DEMOVD68 D298
DEMOVD284 D300
M16
1298
DEMOVD70 D298
DEMOVD286 D300
M14 M15 M16
1317
M23
D66 A/S 1261 1517 2904 A/S
D298 A/S 1332 1358 1398 1407 1446 B/D 1261 1280 1299 1323
D282 A/S 1270 1504 B/D 1145
D300 A/S 3122 3135 3148 B/D 1270 1289 1308
D68 A/S 1280 1559 2913 A/S
D298 A/S 1332 1358 1398 1407 1446 B/D 1261 1280 1299 1323
D284 A/S 1289 1546 B/D 1158
D300 A/S 3122 3135 3148 B/D 1270 1289 1308
D70 A/S 1299 1601 2922 A/S
D298 A/S 1332 1358 1398 1407 1446 B/D 1261 1280 1299 1323
D286 A/S 1308 1588 B/D 1171
D300 A/S 3122 3135 3148 B/D 1270 1289 1308
M23 A/S 1707 1897 B/D 1704

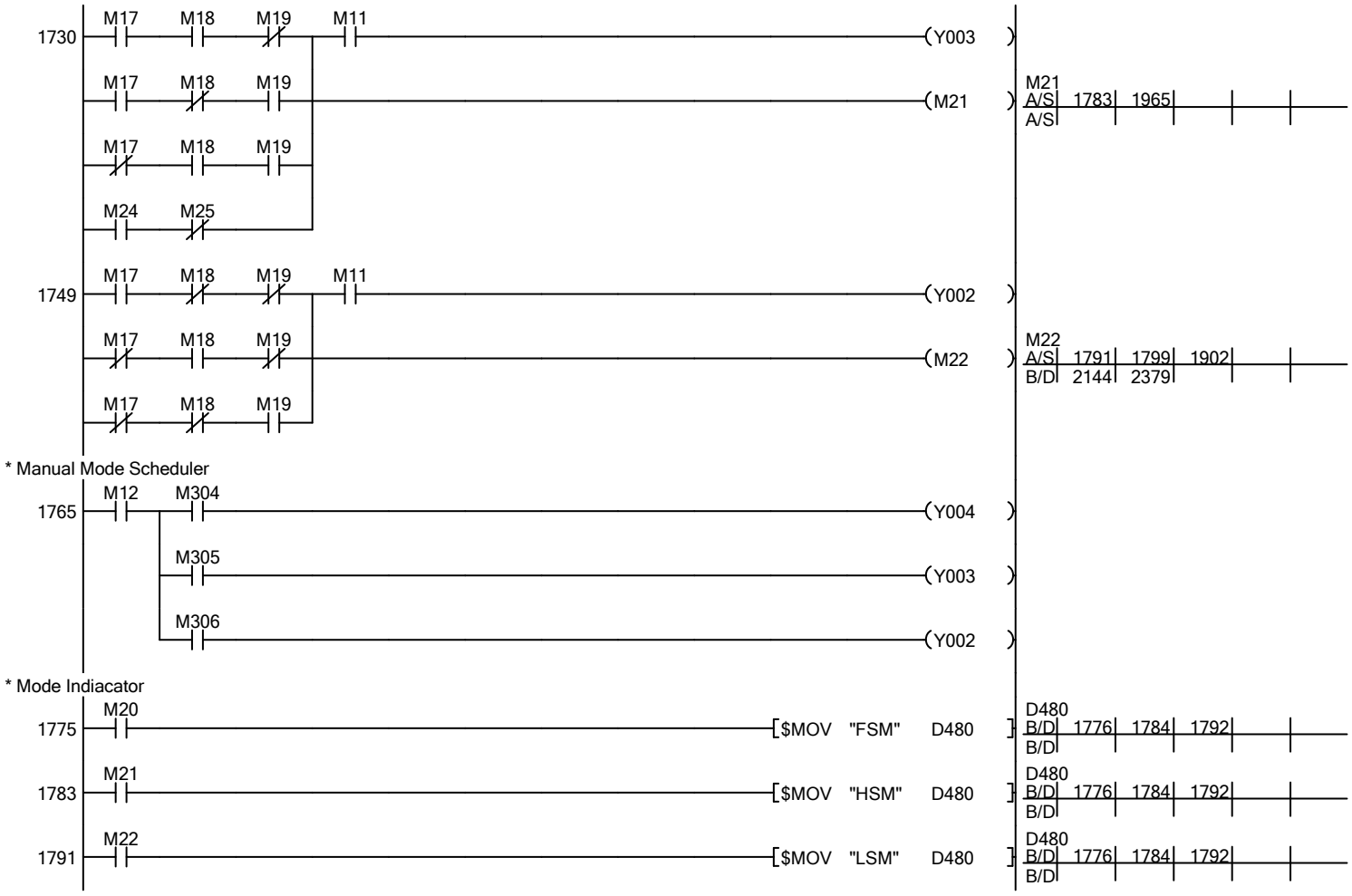

* PO Timer (D302)
1321
T0
D>= D298 K30
D< D298 K30
DESUB K50 D298 D308
DEMUL D308 D400 D310
DESUB D76 D298 D312
DEDIV D310 D312 D314
DEMUL D314 D306 D302
DEMUL D298 E1.9 D308
DESUB E165 D308 D310
DEMUL D310 D400 D312
DESUB D76 D298 D314
DEDIV D312 D314 D316

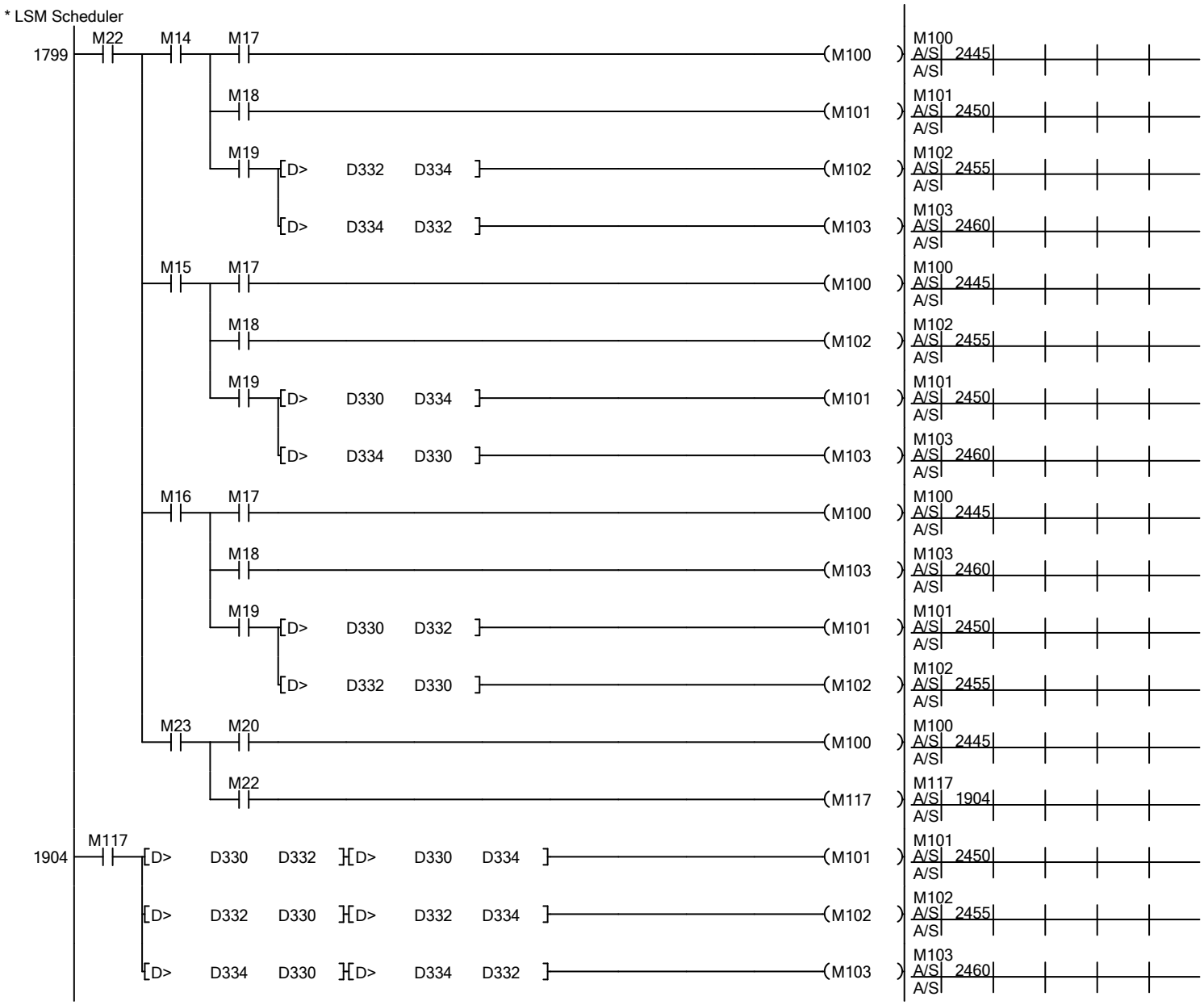

[DEMUL D316 D304 D302]
D316
A/S 1472
B/D 1459 2817 2848 2853 2862
D316
A/S
B/D 2867 2876 2881 2887
D316
A/S 1472
B/D 1459 2817 2848 2853 2862
D316
A/S
B/D 2867 2876 2881 2887
D304
A/S 1472 2670 2703 2736
B/D 77 2679 2712 2745
D302
A/S 1672
B/D 1384 1472 1488 1493
* PO Dispensing
1485
T0
M50
M51
D302
(T5)
T5
A/S 1491
A/S
1491
T5
[DMOV K0 D302]
D302
A/S 1672
B/D 1384 1472 1488 1493
M52
[RST M50]
M50
A/S 1486
B/D 1502

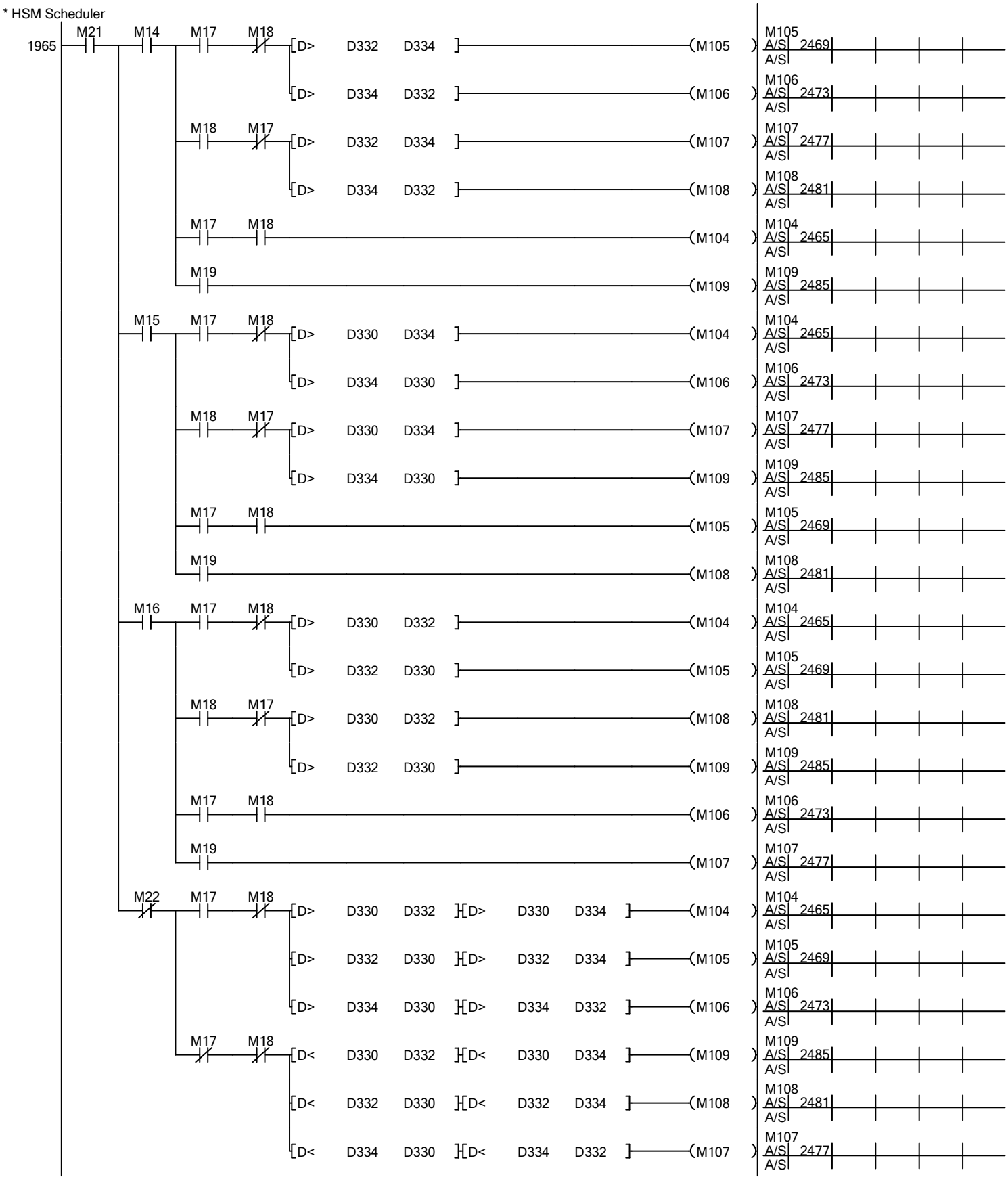

* Tank Scoring
1503
M8000
DEMUL D282 E0.6 D318
D282 A/S 1270 1504 B/D 1145
D318 A/S 1532 B/D 1504
DEMUL D66 E0.4 D320
D66 A/S 1261 1517 2904 A/S
D320 A/S 1532 B/D 1517
M700
DEADD D318 D320 D330
D318 A/S 1532 B/D 1504
D320 A/S 1532 B/D 1517
D330 A/S 1843 1854 1875 1886 1906 B/D 1532 1629
D330 A/S 1915 1926 1946 2031 2042 B/D
D330 A/S 2056 2067 2090 2101 2115 B/D
D330 A/S 2126 2149 2158 2169 2189 B/D
D330 A/S 2212 2221 2232 2252 2322 B/D
D330 A/S 2333 2357 2368 2383 2392 B/D
D330 A/S 2403 2423 B/D
DEMUL D284 E0.6 D322
D284 A/S 1289 1546 B/D 1158
D322 A/S 1574 B/D 1546
DEMUL D68 E0.4 D324
D68 A/S 1280 1559 2913 A/S
D324 A/S 1574 B/D 1559
M701
DEADD D322 D324 D332
D322 A/S 1574 B/D 1546
D324 A/S 1574 B/D 1559
D332 A/S 1811 1822 1875 1886 1906 B/D 1574 1639
D332 A/S 1926 1935 1955 1972 1983 B/D
D332 A/S 1997 2008 2090 2101 2115 B/D
D332 A/S 2126 2149 2169 2178 2198 B/D
D332 A/S 2212 2232 2241 2261 2287 B/D
D332 A/S 2298 2357 2368 2383 2403 B/D
D332 A/S 2412 2432 B/D

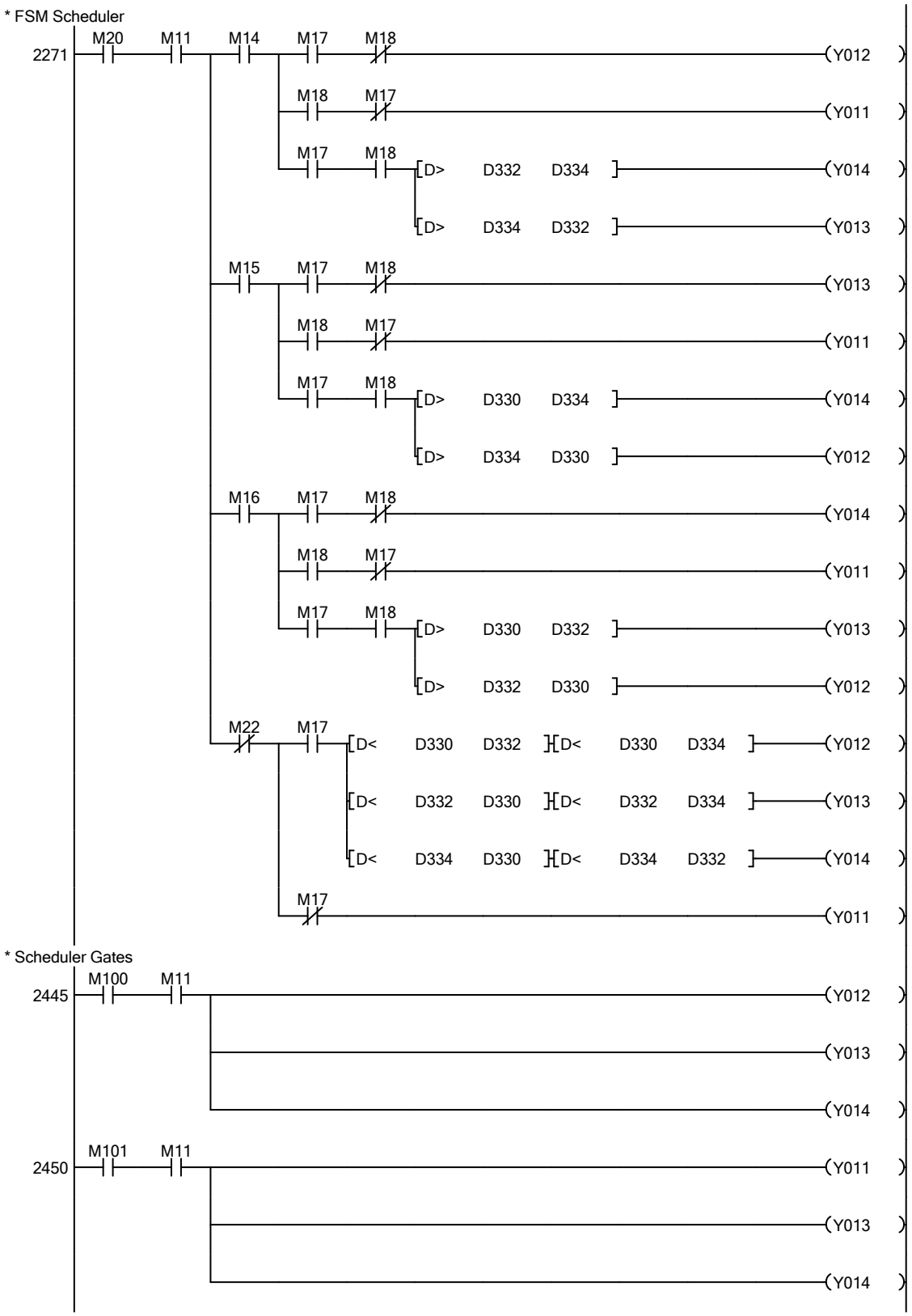

DEMUL D286 E0.6 D326
D286 A/S 1308 1588 B/D 1171
D326 A/S 1615 B/D 1588
DEMUL D70 E0.4 D328
D70 A/S 1299 1601 2922 A/S
D328 A/S 1615 B/D 1601
M702
DEADD D326 D328 D334
D326 A/S 1615 B/D 1588
D328 A/S 1615 B/D 1601
D334 A/S 1811 1822 1843 1854 1915 B/D 1615 1649
D334 A/S 1935 1946 1955 1972 1983 B/D
D334 A/S 1997 2008 2031 2042 2056 B/D
D334 A/S 2067 2158 2178 2189 2198 B/D
D334 A/S 2221 2241 2252 2261 2287 B/D
D334 A/S 2298 2322 2333 2392 2412 B/D
D334 A/S 2423 2432 B/D
1628
M700
DEMOV E0 D330
D330 A/S 1843 1854 1875 1886 1906 B/D 1532 1629
D330 A/S 1915 1926 1946 2031 2042 B/D
D330 A/S 2056 2067 2090 2101 2115 B/D
D330 A/S 2126 2149 2158 2169 2189 B/D
D330 A/S 2212 2221 2232 2252 2322 B/D
D330 A/S 2333 2357 2368 2383 2392 B/D
D330 A/S 2403 2423 B/D
1638
M701
DEMOV E0 D332
D332 A/S 1811 1822 1875 1886 1906 B/D 1574 1639
D332 A/S 1926 1935 1955 1972 1983 B/D
D332 A/S 1997 2008 2090 2101 2115 B/D
D332 A/S 2126 2149 2169 2178 2198 B/D
D332 A/S 2212 2232 2241 2261 2287 B/D
D332 A/S 2298 2357 2368 2383 2403 B/D
D332 A/S 2412 2432 B/D

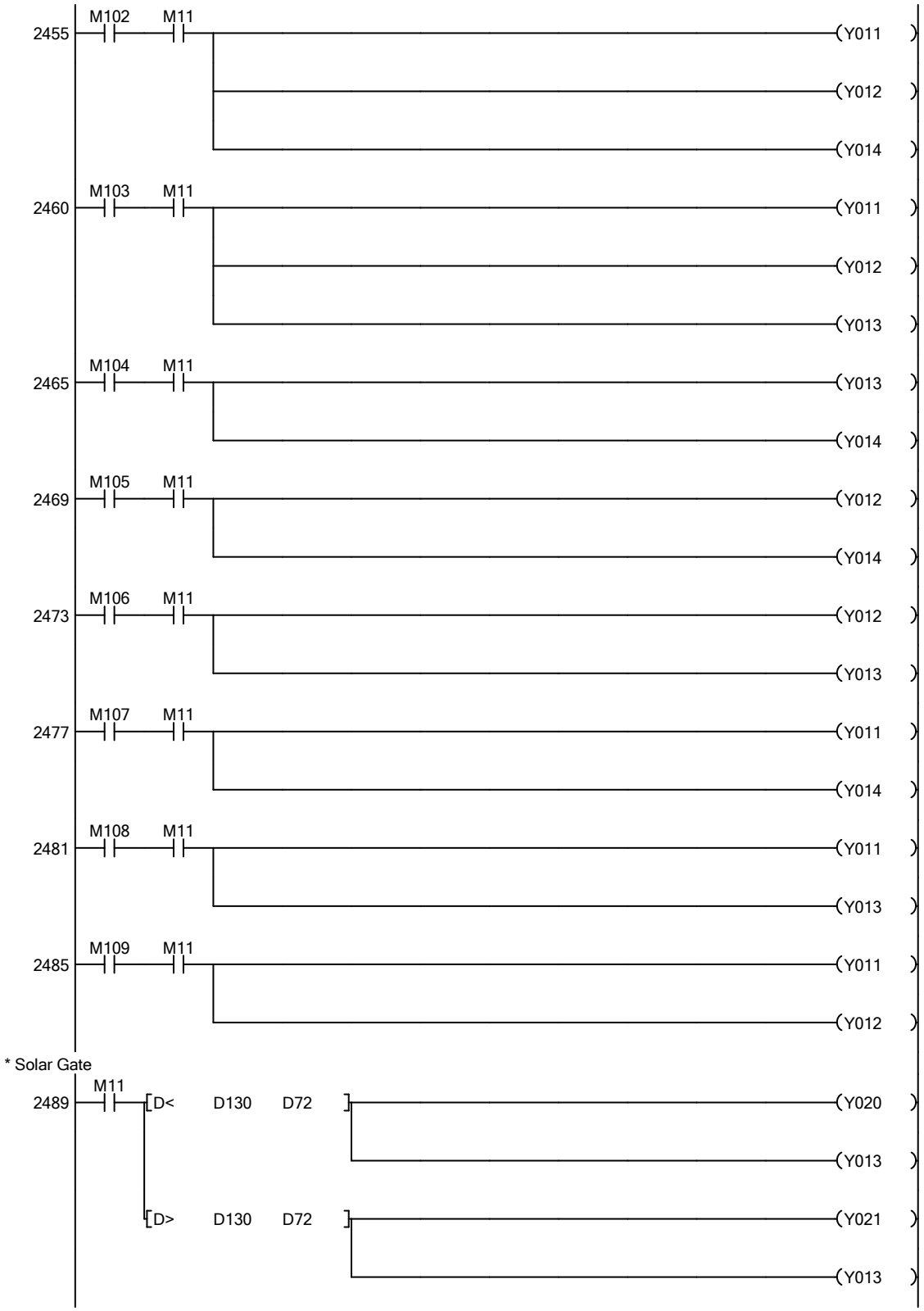

1648
M702
DEMOVE0 D334
D334 A/S 1811 1822 1843 1854 1915 B/D 1615 1649
D334 A/S 1935 1946 1955 1972 1983 B/D
D334 A/S 1997 2008 2031 2042 2056 B/D
D334 A/S 2067 2158 2178 2189 2198 B/D
D334 A/S 2221 2241 2252 2261 2287 B/D
D334 A/S 2298 2322 2333 2392 2412 B/D
D334 A/S 2423 2432 B/D
* Need Scheduler
1658
T1
D< D64 K41
M13
M17
M17 A/S 1711 1718 1730 1733 1749 B/D 1708 1720 1737 1752 1756
M17 A/S 1803 1835 1867 1969 2019 B/D 1995 2054 2113 2209 2281
M17 A/S 2028 2078 2087 2137 2146 B/D 2316 2351 2443
M17 A/S 2276 2284 2311 2319 2346 B/D
M17 A/S 2354 2381 B/D
D> D302 K0
M18
M18 A/S 1712 1721 1731 1738 1753 B/D 1709 1719 1734 1750 1757
M18 A/S 1806 1838 1870 1994 2020 B/D 1970 2029 2088 2147 2210
M18 A/S 2053 2079 2112 2138 2280 B/D 2277 2312 2347
M18 A/S 2285 2315 2320 2350 2355 B/D
D< D122 K90
M19
M19 A/S 1693 1713 1735 1739 1758 B/D 1732 1751 1754
M19 A/S 1809 1841 1873 2023 2082 B/D
M19 A/S 2141 B/D
* Scheduler Cases
1693
M19
D< D122 K70
M23
M24
M24 A/S 1717 1741 B/D 1715
M23 M17 M18
M25
M25 A/S 1714 B/D 1742
* Mode Scheduler
1711
M17 M18 M19 M11
Y004
M25 M24
M20
M20 A/S 1775 1899 2271 A/S
M24 M17 M18
M17 M18

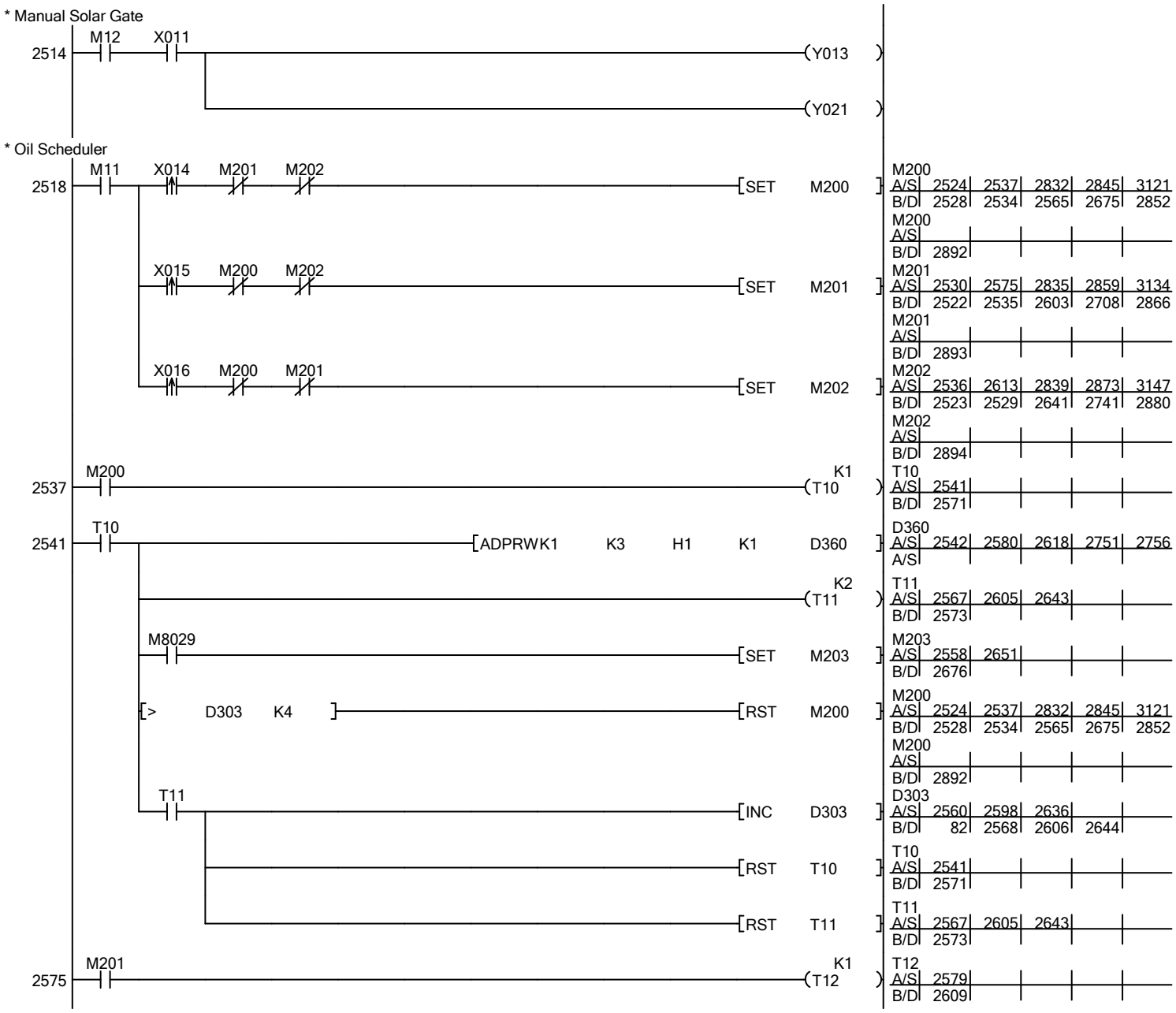

1730
M17 M18 M19 M11
Y003
M17 M18 M19
M21
M21
A/S 1783 1965
A/S
M17 M18 M19
M24 M25
1749
M17 M18 M19 M11
Y002
M17 M18 M19
M22
M22
A/S 1791 1799 1902
B/D 2144 2379
M17 M18 M19
* Manual Mode Scheduler
1765
M12 M304
Y004
M305
Y003
M306
Y002
* Mode Indiacator
1775
M20
[$MOV "FSM" D480]
D480
B/D 1776 1784 1792
B/D
1783
M21
[$MOV "HSM" D480]
D480
B/D 1776 1784 1792
B/D
1791
M22
[$MOV "LSM" D480]
D480
B/D 1776 1784 1792
B/D

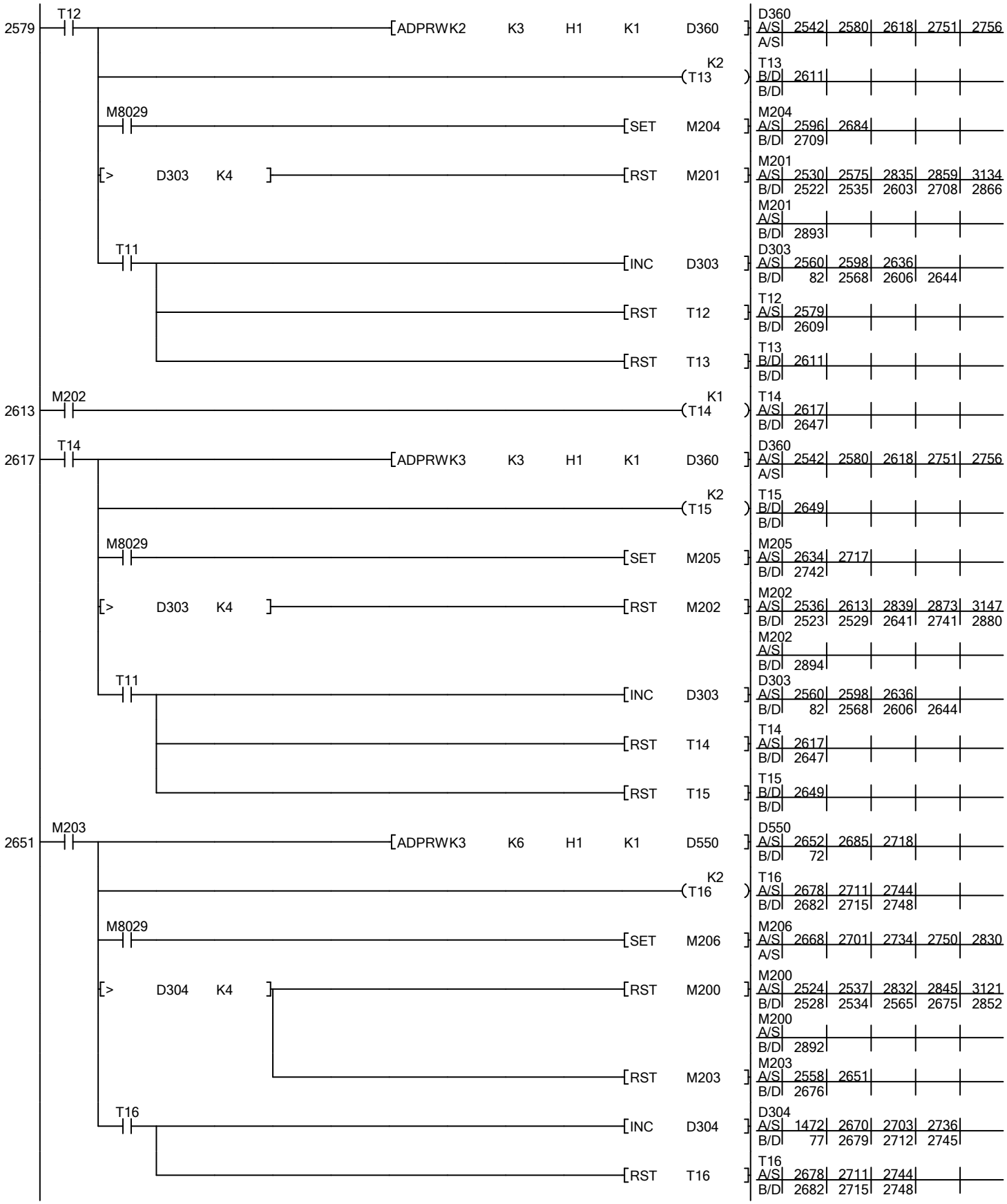

* LSM Scheduler
1799
M22
M14
M17
M100
M100 A/S 2445 A/S
M18
M101
M101 A/S 2450 A/S
M19
D> D332 D334
M102
M102 A/S 2455 A/S
D> D334 D332
M103
M103 A/S 2460 A/S
M15
M17
M100
M100 A/S 2445 A/S
M18
M102
M102 A/S 2455 A/S
M19
D> D330 D334
M101
M101 A/S 2450 A/S
D> D334 D330
M103
M103 A/S 2460 A/S
M16
M17
M100
M100 A/S 2445 A/S
M18
M103
M103 A/S 2460 A/S
M19
D> D330 D332
M101
M101 A/S 2450 A/S
D> D332 D330
M102
M102 A/S 2455 A/S
M23
M20
M100
M100 A/S 2445 A/S
M22
M117
M117 A/S 1904 A/S
1904
M117
D> D330 D332 D> D330 D334
M101
M101 A/S 2450 A/S
D> D332 D330 D> D332 D334
M102
M102 A/S 2455 A/S
D> D334 D330 D> D334 D332
M103
M103 A/S 2460 A/S

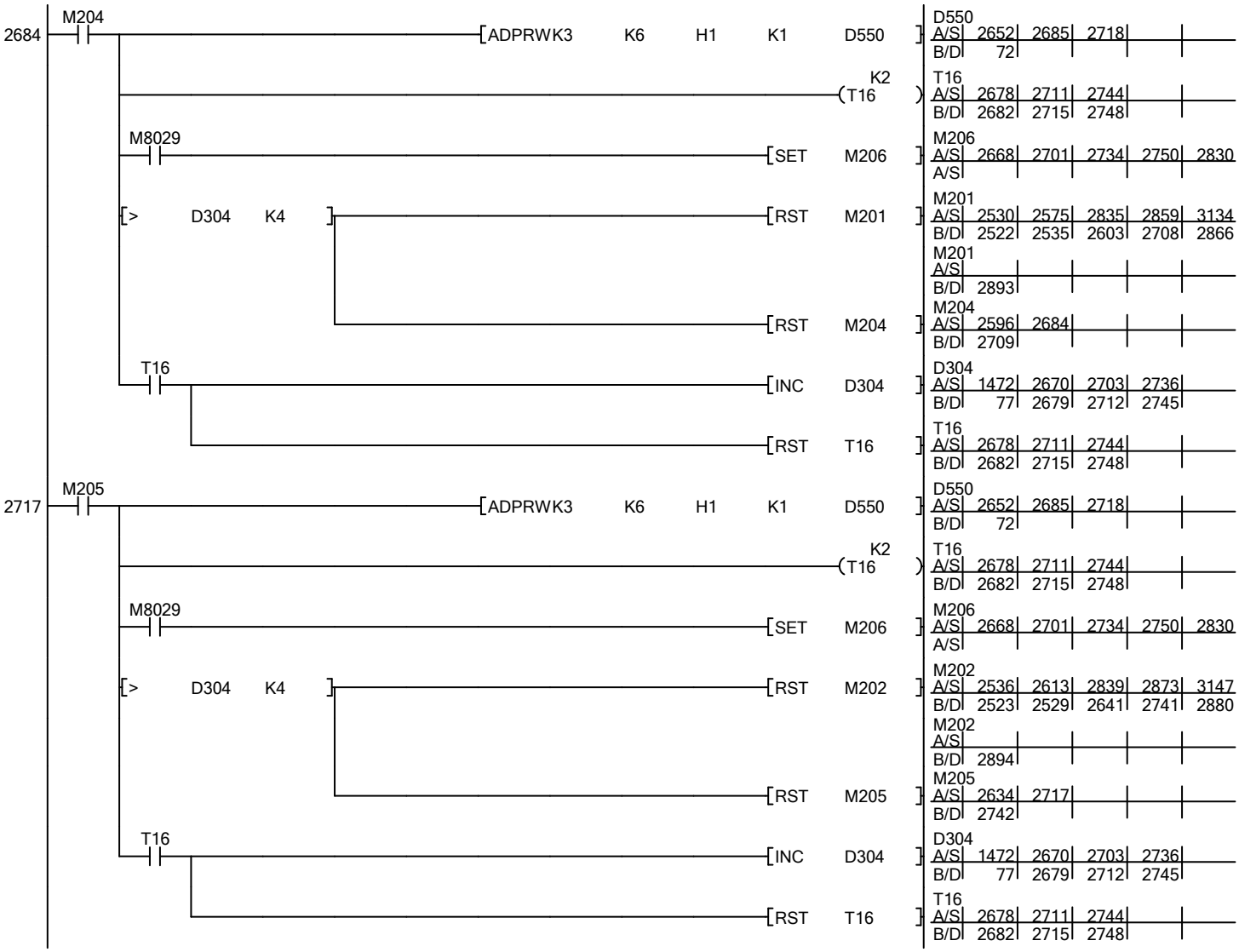

* HSM Scheduler
1965
M21 M14 M17 M18 [D> D332 D334] (M105)
[D> D334 D332] (M106)
M18 M17 [D> D332 D334] (M107)
[D> D334 D332] (M108)
M17 M18 (M104)
M19 (M109)
M15 M17 M18 [D> D330 D334] (M104)
[D> D334 D330] (M106)
M18 M17 [D> D330 D334] (M107)
[D> D334 D330] (M109)
M17 M18 (M105)
M19 (M108)
M16 M17 M18 [D> D330 D332] (M104)
[D> D332 D330] (M105)
M18 M17 [D> D330 D332] (M108)
[D> D332 D330] (M109)
M17 M18 (M106)
M19 (M107)
M22 M17 M18 [D> D330 D332][D> D330 D334] (M104)
[D> D332 D330][D> D332 D334] (M105)
[D> D334 D330][D> D334 D332] (M106)
M17 M18 [D< D330 D332][D< D330 D334] (M109)
[D< D332 D330][D< D332 D334] (M108)
[D< D334 D330][D< D334 D332] (M107)
M105 A/S 2469 A/S
M106 A/S 2473 A/S
M107 A/S 2477 A/S
M108 A/S 2481 A/S
M104 A/S 2465 A/S
M109 A/S 2485 A/S

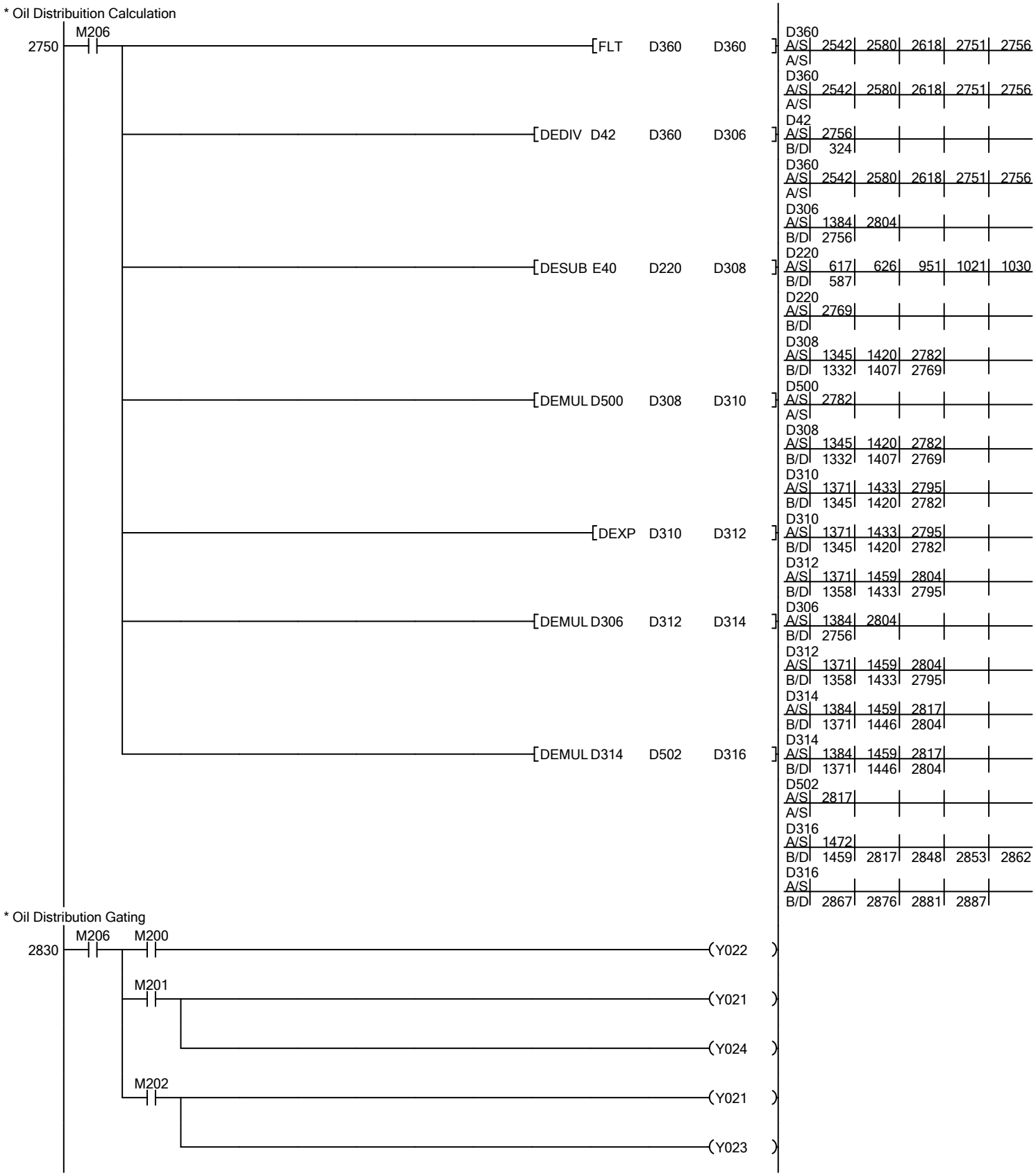

* FSM Scheduler
2271
M20 M11 M14 M17 M18 (Y012)
M18 M17 (Y011)
M17 M18 [D> D332 D334] (Y014)
[D> D334 D332] (Y013)
M15 M17 M18 (Y013)
M18 M17 (Y011)
M17 M18 [D> D330 D334] (Y014)
[D> D334 D330] (Y012)
M16 M17 M18 (Y014)
M18 M17 (Y011)
M17 M18 [D> D330 D332] (Y013)
[D> D332 D330] (Y012)
M22 M17 [D< D330 D332][D< D330 D334] (Y012)
[D< D332 D330][D< D332 D334] (Y013)
[D< D334 D330][D< D334 D332] (Y014)
M17 (Y011)
* Scheduler Gates
2445
M100 M11 (Y012)
(Y013)
(Y014)
2450
M101 M11 (Y011)
(Y013)
(Y014)

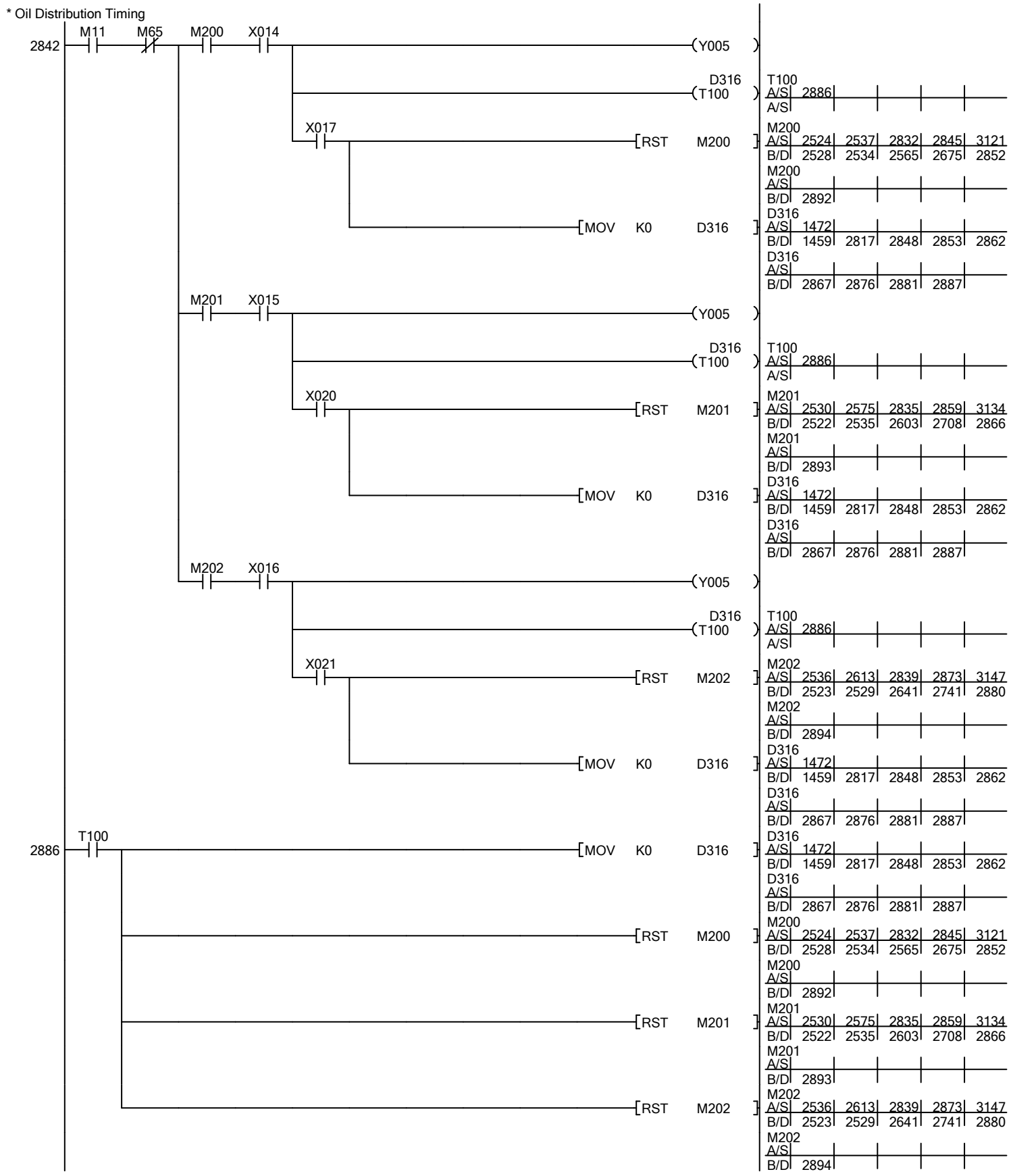

2455
M102
M11
Y011
Y012
Y014
2460
M103
M11
Y011
Y012
Y013
2465
M104
M11
Y013
Y014
2469
M105
M11
Y012
Y014
2473
M106
M11
Y012
Y013
2477
M107
M11
Y011
Y014
2481
M108
M11
Y011
Y013
2485
M109
M11
Y011
Y012
* Solar Gate
2489
M11
D< D130 D72
Y020
Y013
D> D130 D72
Y021
Y013

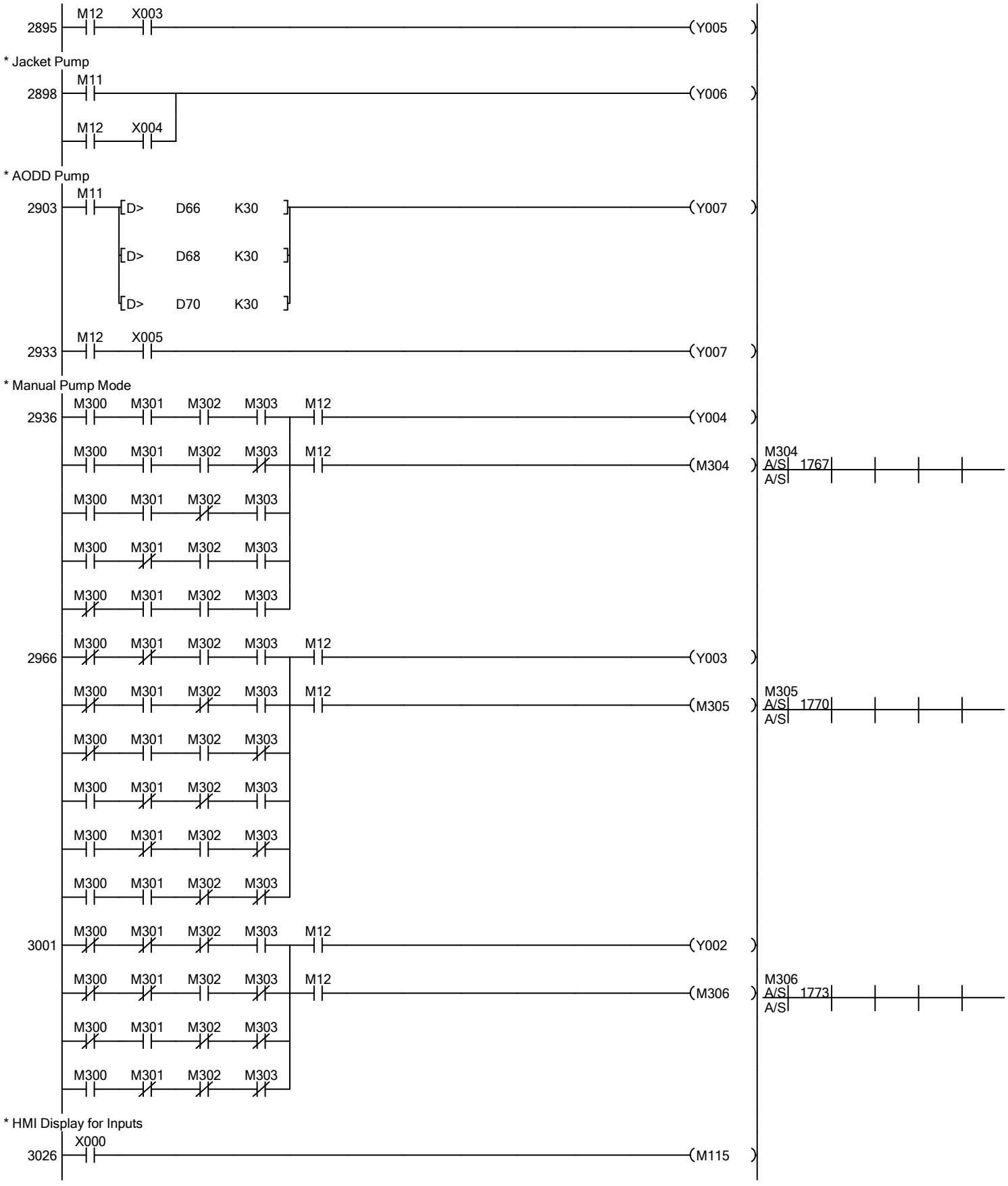

* Manual Solar Gate
2514
M12
X011
Y013
Y021
* Oil Scheduler
2518
M11
X014
M201
M202
SET M200
X015
M200
M202
SET M201
X016
M200
M201
SET M202
2537
M200
K1
T10
2541
T10
ADPRWK1 K3 H1 K1 D360
K2
T11
M8029
SET M203
> D303 K4
RST M200
T11
INC D303
RST T10
RST T11
2575
M201
K1
T12
M200 A/S 2524 2537 2832 2845 3121 B/D 2528 2534 2565 2675 2852
M200 A/S B/D 2892
M201 A/S 2530 2575 2835 2859 3134 B/D 2522 2535 2603 2708 2866
M201 A/S B/D 2893
M202 A/S 2536 2613 2839 2873 3147 B/D 2523 2529 2641 2741 2880
M202 A/S B/D 2894
T10 A/S 2541 B/D 2571
D360 A/S 2542 2580 2618 2751 2756 A/S
T11 A/S 2567 2605 2643 B/D 2573
M203 A/S 2558 2651 B/D 2676
M200 A/S 2524 2537 2832 2845 3121 B/D 2528 2534 2565 2675 2852
M200 A/S B/D 2892
D303 A/S 2560 2598 2636 B/D 82 2568 2606 2644
T10 A/S 2541 B/D 2571
T11 A/S 2567 2605 2643 B/D 2573
T12 A/S 2579 B/D 2609

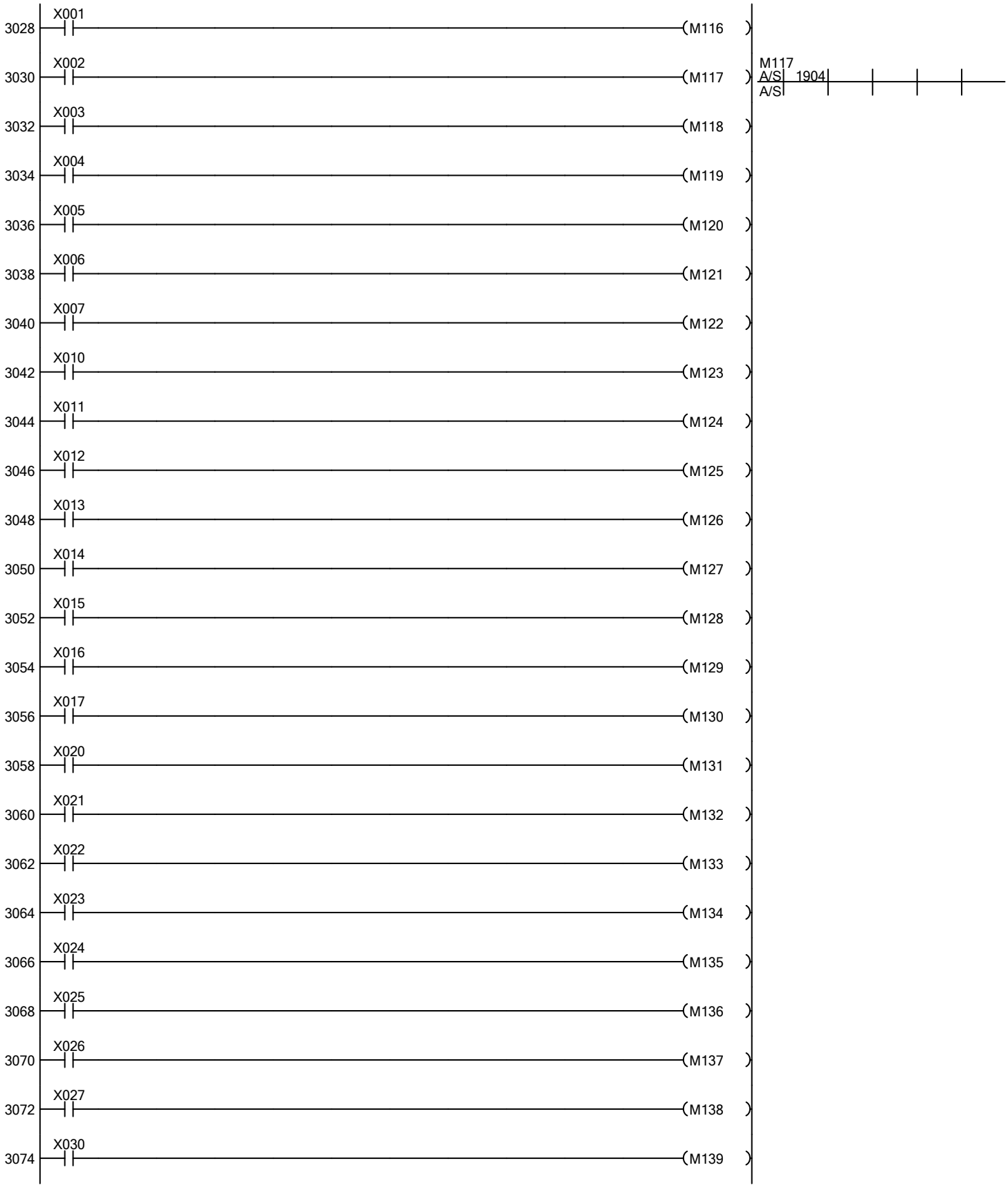

2579 T12 [ADPRW K2 K3 H1 K1 D360]
D360 A/S 2542 2580 2618 2751 2756
(T13 K2)
T13 B/D 2611
M8029 [SET M204]
M204 A/S 2596 2684 B/D 2709
[> D303 K4] [RST M201]
M201 A/S 2530 2575 2835 2859 3134 B/D 2522 2535 2603 2708 2866
M201 A/S B/D 2893
T11 [INC D303]
D303 A/S 2560 2598 2636 B/D 82 2568 2606 2644
[RST T12]
T12 A/S 2579 B/D 2609
[RST T13]
T13 B/D 2611
2613 M202 (T14 K1)
T14 A/S 2617 B/D 2647
2617 T14 [ADPRW K3 K3 H1 K1 D360]
D360 A/S 2542 2580 2618 2751 2756
(T15 K2)
T15 B/D 2649
M8029 [SET M205]
M205 A/S 2634 2717 B/D 2742
[> D303 K4] [RST M202]
M202 A/S 2536 2613 2839 2873 3147 B/D 2523 2529 2641 2741 2880
M202 A/S B/D 2894
T11 [INC D303]
D303 A/S 2560 2598 2636 B/D 82 2568 2606 2644
[RST T14]
T14 A/S 2617 B/D 2647
[RST T15]
T15 B/D 2649
2651 M203 [ADPRW K3 K6 H1 K1 D550]
D550 A/S 2652 2685 2718 B/D 72
(T16 K2)
T16 A/S 2678 2711 2744 B/D 2682 2715 2748
M8029 [SET M206]
M206 A/S 2668 2701 2734 2750 2830
[> D304 K4] [RST M200]
M200 A/S 2524 2537 2832 2845 3121 B/D 2528 2534 2565 2675 2852
M200 A/S B/D 2892
[RST M203]
M203 A/S 2558 2651 B/D 2676
T16 [INC D304]
D304 A/S 1472 2670 2703 2736 B/D 77 2679 2712 2745
[RST T16]
T16 A/S 2678 2711 2744 B/D 2682 2715 2748

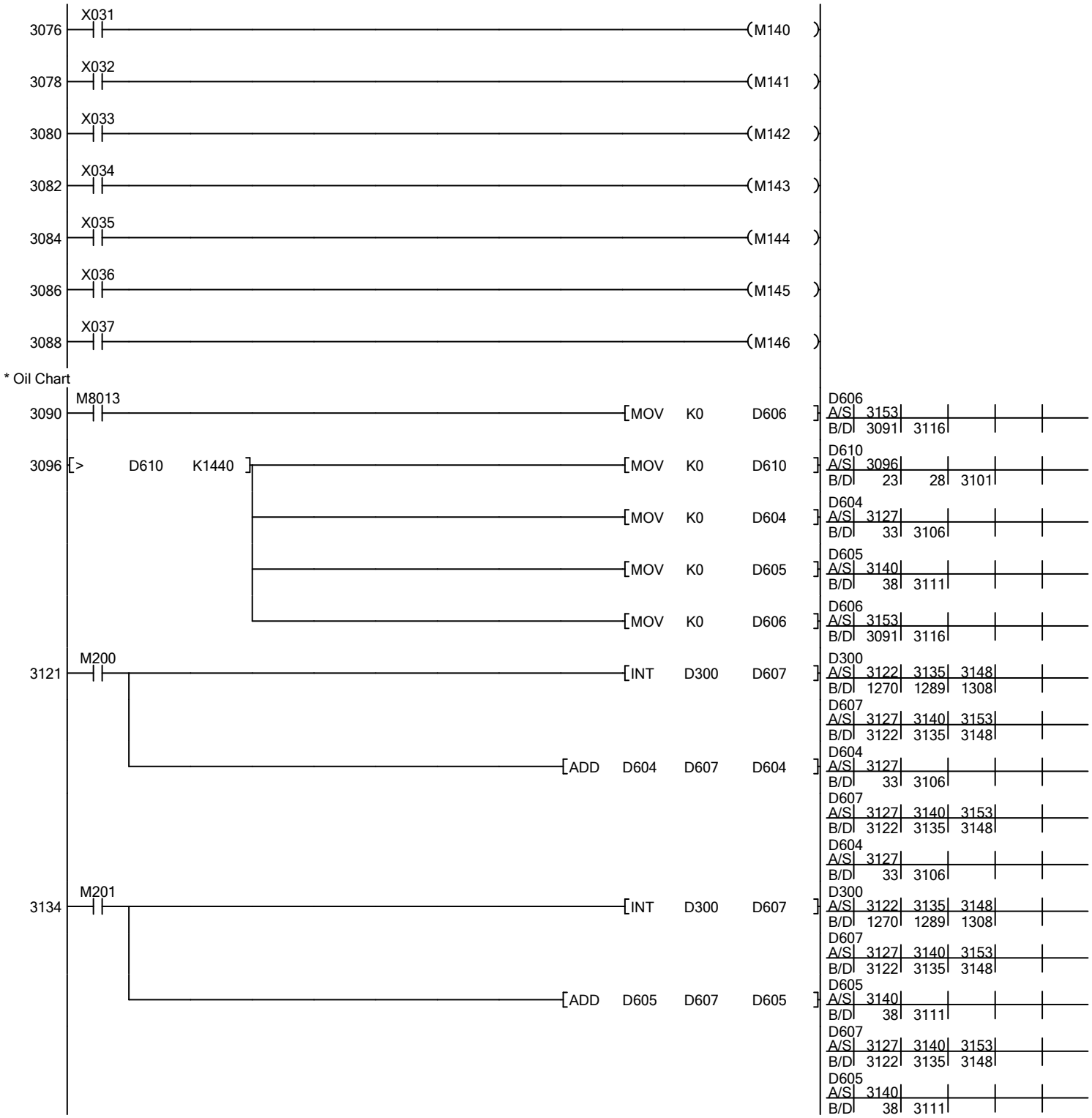

2684
M204
ADPRW K3 K6 H1 K1 D550
K2
T16
M8029
SET M206
> D304 K4
RST M201
RST M204
T16
INC D304
RST T16
2717
M205
ADPRW K3 K6 H1 K1 D550
K2
T16
M8029
SET M206
> D304 K4
RST M202
RST M205
T16
INC D304
RST T16
D550 A/S 2652 2685 2718 B/D 72
T16 A/S 2678 2711 2744 B/D 2682 2715 2748
M206 A/S 2668 2701 2734 2750 2830 A/S
M201 A/S 2530 2575 2835 2859 3134 B/D 2522 2535 2603 2708 2866
M201 A/S B/D 2893
M204 A/S 2596 2684 B/D 2709
D304 A/S 1472 2670 2703 2736 B/D 77 2679 2712 2745
T16 A/S 2678 2711 2744 B/D 2682 2715 2748
D550 A/S 2652 2685 2718 B/D 72
T16 A/S 2678 2711 2744 B/D 2682 2715 2748
M206 A/S 2668 2701 2734 2750 2830 A/S
M202 A/S 2536 2613 2839 2873 3147 B/D 2523 2529 2641 2741 2880
M202 A/S B/D 2894
M205 A/S 2634 2717 B/D 2742
D304 A/S 1472 2670 2703 2736 B/D 77 2679 2712 2745
T16 A/S 2678 2711 2744 B/D 2682 2715 2748

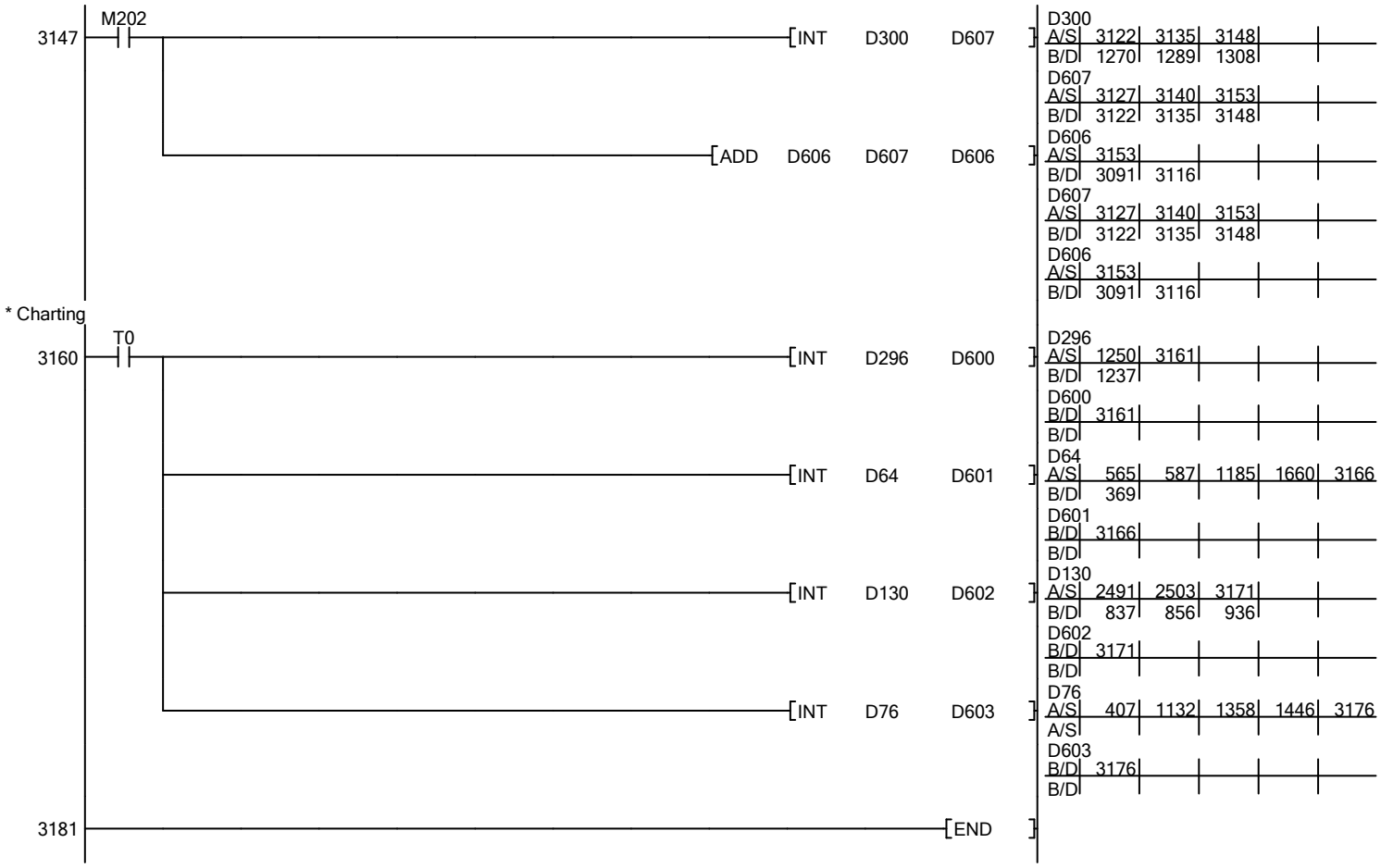

* Oil Distribuition Calculation
2750
M206
FLT D360 D360
DEDIV D42 D360 D306
DESUB E40 D220 D308
DEMUL D500 D308 D310
DEXP D310 D312
DEMUL D306 D312 D314
DEMUL D314 D502 D316
D360 A/S 2542 2580 2618 2751 2756 A/S
D360 A/S 2542 2580 2618 2751 2756 A/S
D42 A/S 2756 B/D 324
D360 A/S 2542 2580 2618 2751 2756 A/S
D306 A/S 1384 2804 B/D 2756
D220 A/S 617 626 951 1021 1030 B/D 587
D220 A/S 2769 B/D
D308 A/S 1345 1420 2782 B/D 1332 1407 2769
D500 A/S 2782 A/S
D308 A/S 1345 1420 2782 B/D 1332 1407 2769
D310 A/S 1371 1433 2795 B/D 1345 1420 2782
D310 A/S 1371 1433 2795 B/D 1345 1420 2782
D312 A/S 1371 1459 2804 B/D 1358 1433 2795
D306 A/S 1384 2804 B/D 2756
D312 A/S 1371 1459 2804 B/D 1358 1433 2795
D314 A/S 1384 1459 2817 B/D 1371 1446 2804
D314 A/S 1384 1459 2817 B/D 1371 1446 2804
D502 A/S 2817 A/S
D316 A/S 1472 B/D 1459 2817 2848 2853 2862
D316 A/S B/D 2867 2876 2881 2887
* Oil Distribution Gating
2830
M206 M200
Y022
M201
Y021
Y024
M202
Y021
Y023


* Oil Distribution Timing
2842
M11 M65 M200 X014
Y005
D316
T100
T100
A/S 2886
A/S
X017
RST M200
M200
A/S 2524 2537 2832 2845 3121
B/D 2528 2534 2565 2675 2852
M200
A/S
B/D 2892
MOV K0 D316
D316
A/S 1472
B/D 1459 2817 2848 2853 2862
D316
A/S
B/D 2867 2876 2881 2887
M201 X015
Y005
D316
T100
T100
A/S 2886
A/S
X020
RST M201
M201
A/S 2530 2575 2835 2859 3134
B/D 2522 2535 2603 2708 2866
M201
A/S
B/D 2893
MOV K0 D316
D316
A/S 1472
B/D 1459 2817 2848 2853 2862
D316
A/S
B/D 2867 2876 2881 2887
M202 X016
Y005
D316
T100
T100
A/S 2886
A/S
X021
RST M202
M202
A/S 2536 2613 2839 2873 3147
B/D 2523 2529 2641 2741 2880
M202
A/S
B/D 2894
MOV K0 D316
D316
A/S 1472
B/D 1459 2817 2848 2853 2862
D316
A/S
B/D 2867 2876 2881 2887
2886
T100
MOV K0 D316
D316
A/S 1472
B/D 1459 2817 2848 2853 2862
D316
A/S
B/D 2867 2876 2881 2887
RST M200
M200
A/S 2524 2537 2832 2845 3121
B/D 2528 2534 2565 2675 2852
M200
A/S
B/D 2892
RST M201
M201
A/S 2530 2575 2835 2859 3134
B/D 2522 2535 2603 2708 2866
M201
A/S
B/D 2893
RST M202
M202
A/S 2536 2613 2839 2873 3147
B/D 2523 2529 2641 2741 2880
M202
A/S
B/D 2894


2895
M12
X003
Y005
* Jacket Pump
2898
M11
M12
X004
Y006
* AODD Pump
2903
M11
D> D66 K30
D> D68 K30
D> D70 K30
Y007
2933
M12
X005
Y007
* Manual Pump Mode
2936
M300
M301
M302
M303
M12
Y004
M304
A/S 1767
A/S
2966
Y003
M305
A/S 1770
3001
Y002
M306
A/S 1773
* HMI Display for Inputs
3026
X000
M115


3028 X001 M116
3030 X002 M117
3032 X003 M118
3034 X004 M119
3036 X005 M120
3038 X006 M121
3040 X007 M122
3042 X010 M123
3044 X011 M124
3046 X012 M125
3048 X013 M126
3050 X014 M127
3052 X015 M128
3054 X016 M129
3056 X017 M130
3058 X020 M131
3060 X021 M132
3062 X022 M133
3064 X023 M134
3066 X024 M135
3068 X025 M136
3070 X026 M137
3072 X027 M138
3074 X030 M139
M117
A/S 1904
A/S


3076 X031 M140
3078 X032 M141
3080 X033 M142
3082 X034 M143
3084 X035 M144
3086 X036 M145
3088 X037 M146
* Oil Chart
3090 M8013 MOV K0 D606
3096 > D610 K1440 MOV K0 D610
MOV K0 D604
MOV K0 D605
MOV K0 D606
3121 M200 INT D300 D607
ADD D604 D607 D604
3134 M201 INT D300 D607
ADD D605 D607 D605
D606 A/S 3153 B/D 3091 3116
D610 A/S 3096 B/D 23 28 3101
D604 A/S 3127 B/D 33 3106
D605 A/S 3140 B/D 38 3111
D606 A/S 3153 B/D 3091 3116
D300 A/S 3122 3135 3148 B/D 1270 1289 1308
D607 A/S 3127 3140 3153 B/D 3122 3135 3148
D604 A/S 3127 B/D 33 3106
D607 A/S 3127 3140 3153 B/D 3122 3135 3148
D604 A/S 3127 B/D 33 3106
D300 A/S 3122 3135 3148 B/D 1270 1289 1308
D607 A/S 3127 3140 3153 B/D 3122 3135 3148
D605 A/S 3140 B/D 38 3111
D607 A/S 3127 3140 3153 B/D 3122 3135 3148
D605 A/S 3140 B/D 38 3111


3147
M202
INT D300 D607
ADD D606 D607 D606
* Charting
3160
T0
INT D296 D600
INT D64 D601
INT D130 D602
INT D76 D603
3181
END
D300 A/S 3122 3135 3148 B/D 1270 1289 1308
D607 A/S 3127 3140 3153 B/D 3122 3135 3148
D606 A/S 3153 B/D 3091 3116
D607 A/S 3127 3140 3153 B/D 3122 3135 3148
D606 A/S 3153 B/D 3091 3116
D296 A/S 1250 3161 B/D 1237
D600 B/D 3161 B/D
D64 A/S 565 587 1185 1660 3166 B/D 369
D601 B/D 3166 B/D
D130 A/S 2491 2503 3171 B/D 837 856 936
D602 B/D 3171 B/D
D76 A/S 407 1132 1358 1446 3176 A/S
D603 B/D 3176 B/D

Position	Device	Setting Value
(1)	T0	K10
(4)	T1	K30
(7)	T2	K10
(10)	T3	K50
(13)	T4	K110
(1488)	T5	D302
(2538)	T10	K1
(2553)	T11	K2
(2576)	T12	K1
(2591)	T13	K2
(2614)	T14	K1
(2629)	T15	K2
(2663)	T16	K2
(2696)	T16	K2
(2729)	T16	K2
(2848)	T100	D316
(2862)	T100	D316
(2876)	T100	D316

Data Name : Device List

Find In:Main_2
Find What:Used Device (Contact & Coil)
Print Range:Whole Range

*:in use (times): the number of coil uses

Device	Contact	Coil	Parameter	Comment
M0	*	*(1)		
M1	*	*(1)		
M2	*	*(1)		
M3	*	*(1)		
M4	*	*(1)		
M5	*	*(1)		
M6	*	*(1)		
M7	*	*(1)		
M8	*	*(1)		
M9	*	*(1)		
M10	*	*(1)		
M11	*	*(1)		
M12	*	*(1)		
M13	*	*(1)		
M17	*	*(1)		
M18	*	*(1)		
M19	*	*(1)		
M20	*	*(1)		
M21	*	*(1)		
M22	*	*(1)		
M23	*	*(1)		
M24	*	*(1)		
M25	*	*(1)		
M50	*	*(1)		
M100	*	*(4)		
M101	*	*(4)		
M102	*	*(4)		
M103	*	*(4)		
M104	*	*(4)		
M105	*	*(4)		
M106	*	*(4)		
M107	*	*(4)		
M108	*	*(4)		
M109	*	*(4)		
M117	*	*(2)		
M200	*	*(5)		
M201	*	*(5)		
M202	*	*(5)		
M203	*	*(2)		
M204	*	*(2)		
M205	*	*(2)		
M206	*	*(3)		

Data Name : Device List

Find In:Main_2
Find What:Used Device (Contact & Coil)
Print Range:Whole Range

*:in use (times): the number of coil uses

Device	Contact	Coil	Parameter	Comment
M304	*	*(1)		
M305	*	*(1)		
M306	*	*(1)		
D0	*	*(2)		
D1	*	*(1)		
D2	*	*(2)		
D12	*	*(1)		
D13	*	*(1)		
D20	*	*(2)		
D21	*	*(1)		
D22	*	*(1)		
D23	*	*(1)		
D42	*	*(1)		
D43	*	*(1)		
D44	*	*(2)		
D45	*	*(2)		
D64	*	*(1)		
D65	*	*(1)		
D100	*	*(1)		
D101	*	*(1)		
D102	*	*(1)		
D103	*	*(1)		
D104	*	*(1)		
D105	*	*(1)		
D106	*	*(1)		
D107	*	*(1)		
D108	*	*(1)		
D109	*	*(1)		
D110	*	*(1)		
D111	*	*(1)		
D112	*	*(1)		
D113	*	*(1)		
D114	*	*(2)		
D115	*	*(2)		
D116	*	*(2)		
D117	*	*(2)		
D118	*	*(2)		
D119	*	*(2)		
D120	*	*(2)		
D121	*	*(2)		
D122	*	*(1)		
D123	*	*(1)		

Data Name : Device List

Find In:Main_2
Find What:Used Device (Contact & Coil)
Print Range:Whole Range

*:in use (times): the number of coil uses

Device	Contact	Coil	Parameter	Comment
D130	*	*(3)		
D131	*	*(3)		
D132	*	*(1)		
D133	*	*(1)		
D134	*	*(1)		
D135	*	*(1)		
D136	*	*(1)		
D137	*	*(1)		
D138	*	*(1)		
D139	*	*(1)		
D200	*	*(1)		
D220	*	*(1)		
D221	*	*(1)		
D280	*	*(1)		
D281	*	*(1)		
D282	*	*(1)		
D283	*	*(1)		
D284	*	*(1)		
D285	*	*(1)		
D286	*	*(1)		
D287	*	*(1)		
D288	*	*(1)		
D289	*	*(1)		
D290	*	*(1)		
D291	*	*(1)		
D292	*	*(1)		
D293	*	*(1)		
D294	*	*(1)		
D295	*	*(1)		
D296	*	*(1)		
D297	*	*(1)		
D298	*	*(3)		
D299	*	*(3)		
D300	*	*(3)		
D301	*	*(3)		
D302	*	*(3)		
D303	*	*(7)		
D304	*	*(4)		
D306	*	*(1)		
D307	*	*(1)		
D308	*	*(3)		
D309	*	*(3)		

Data Name : Device List

Find In:Main_2
Find What:Used Device (Contact & Coil)
Print Range:Whole Range

*:in use (times): the number of coil uses

Device	Contact	Coil	Parameter	Comment
D310	*	*(3)		
D311	*	*(3)		
D312	*	*(3)		
D313	*	*(3)		
D314	*	*(3)		
D315	*	*(3)		
D316	*	*(6)		
D317	*	*(2)		
D318	*	*(1)		
D319	*	*(1)		
D320	*	*(1)		
D321	*	*(1)		
D322	*	*(1)		
D323	*	*(1)		
D324	*	*(1)		
D325	*	*(1)		
D326	*	*(1)		
D327	*	*(1)		
D328	*	*(1)		
D329	*	*(1)		
D330	*	*(2)		
D331	*	*(2)		
D332	*	*(2)		
D333	*	*(2)		
D334	*	*(2)		
D335	*	*(2)		
D350	*	*(8)		
D360	*	*(1)		
D361	*	*(1)		
D550	*	*(1)		
D604	*	*(3)		
D605	*	*(3)		
D606	*	*(3)		
D607	*	*(3)		
D610	*	*(3)		
T0	*	*(1)		
T1	*	*(1)		
T2	*	*(2)		
T3	*	*(2)		
T4	*	*(1)		
T5	*	*(1)		
T10	*	*(2)		

Data Name : Device List

Find In:Main_2
Find What:Used Device (Contact & Coil)
Print Range:Whole Range

*:in use (times): the number of coil uses

Device	Contact	Coil	Parameter	Comment
T11	*	*(2)		
T12	*	*(2)		
T14	*	*(2)		
T16	*	*(6)		
T100	*	*(3)		
Z0	*	*(3)		
Z1	*	*(3)		
Z2	*	*(3)		
Z3	*	*(4)		
Z4	*	*(3)		
Z5	*	*(3)		
P0	*	*(1)		
P1	*	*(1)		
P2	*	*(1)		
P3	*	*(1)		
P4	*	*(1)		
P5	*	*(1)		

[Line Statement List]

Step	Line Statement
0	Timers
22	Load Once
43	E-Stops
57	Automation
66	Declarations
134	Analog to Digital
168	ADC Float
310	Flowmeter to RPM
406	Level % Converter
615	Error Threshold
826	Water Temperature Target Logic
949	Heater latching
1121	Manual Heater Control
1131	Height to Mass
1184	Dynamic Upper Thermal Boundary
1260	PO Designation
1321	PO Timer (D302)
1485	PO Dispensing
1503	Tank Scoring
1658	Need Scheduler
1693	Scheduler Cases
1711	Mode Scheduler
1765	Manual Mode Scheduler
1775	Mode Indiacator
1799	LSM Scheduler
1965	HSM Scheduler
2271	FSM Scheduler
2445	Scheduler Gates
2489	Solar Gate
2514	Manual Solar Gate
2518	Oil Scheduler
2750	Oil Distribuition Calculation
2830	Oil Distribution Gating
2842	Oil Distribution Timing
2898	Jacket Pump
2903	AODD Pump
2936	Manual Pump Mode
3026	HMI Display for Inputs
3090	Oil Chart
3160	Charting

[P Statement List]

Device	P Statement
P0	RTD to Temperature
P1	Ultrasonic to Level
P2	Sensor Averaging
P4	Sensor Logging (5 sec)

Data Name : Project Contents List

Workspace Name :
Project Name : Main_2
Title :

Data Name	Last Change	Title
Parameter	9/6/2026 12:12:22 PM	
PLC Parameter	9/6/2026 12:12:22 PM	
Network Parameter	9/6/2026 12:12:22 PM	
CC-Link	9/6/2026 12:12:22 PM	
Global Device Comment	9/6/2026 12:12:23 PM	
Program Setting		
Execution Program		
MAIN	9/6/2026 12:12:23 PM	
MAIN	9/9/2026 3:58:12 PM	
POU		
Program	9/6/2026 12:12:23 PM	
MAIN	9/9/2026 3:58:12 PM	
Local Device Comment		
Device Memory	9/6/2026 12:12:37 PM	
MAIN	9/6/2026 12:12:37 PM	